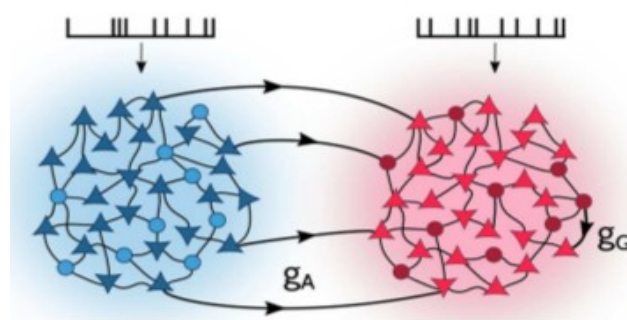
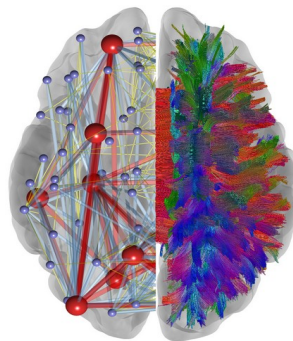
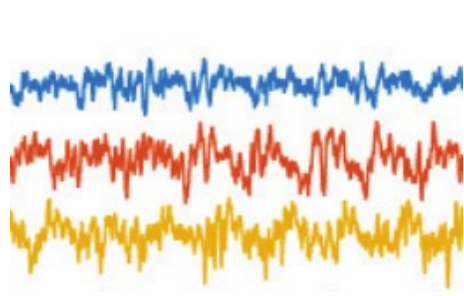
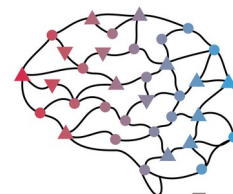




An Introduction to Computational Neuroscience

Fernanda Selingardi Matias

Universidade Federal de Alagoas, Maceió, Brasil



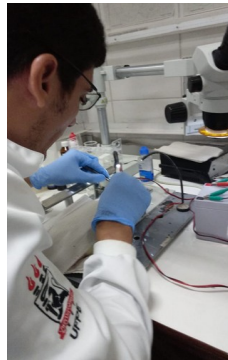
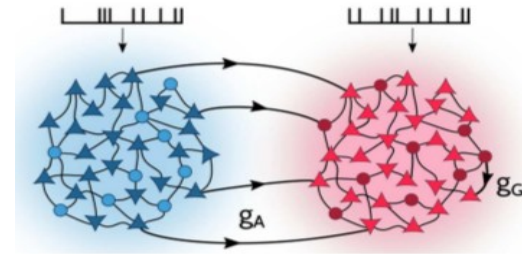
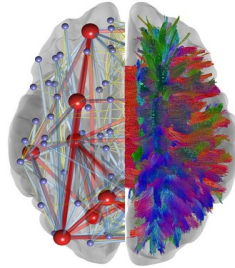
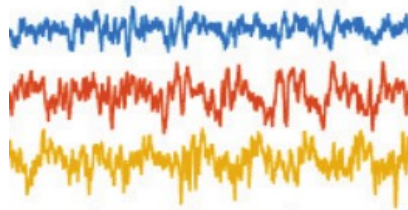
NeuroLab

fernanda@fis.ufal.br

Is the brain a machine that interpret reality?

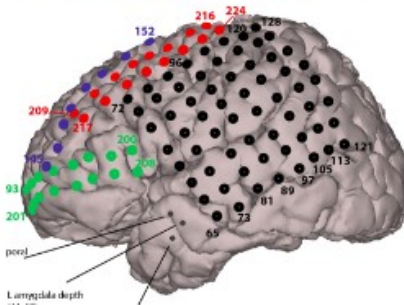
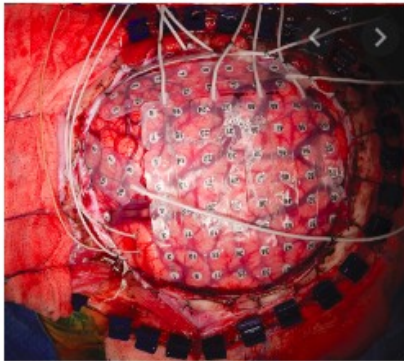
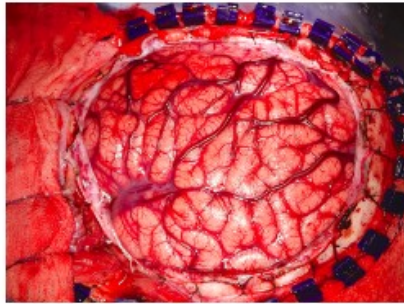


How the brain interprets reality?

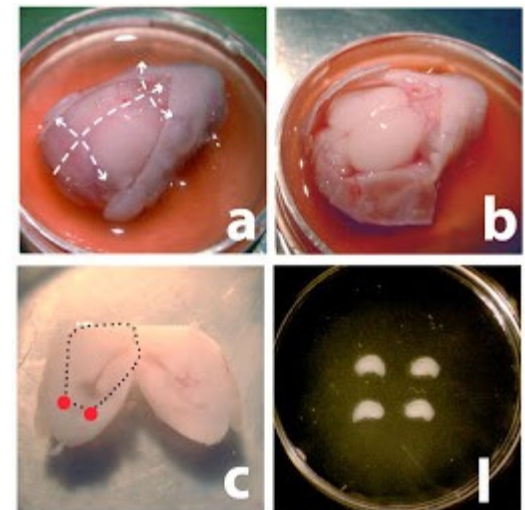


How the brain interprets reality?

in vivo

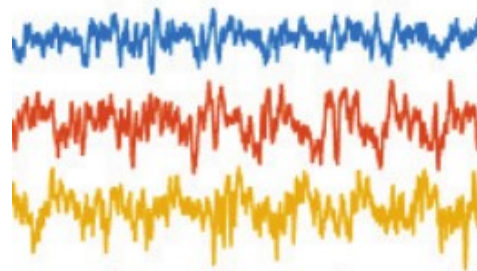
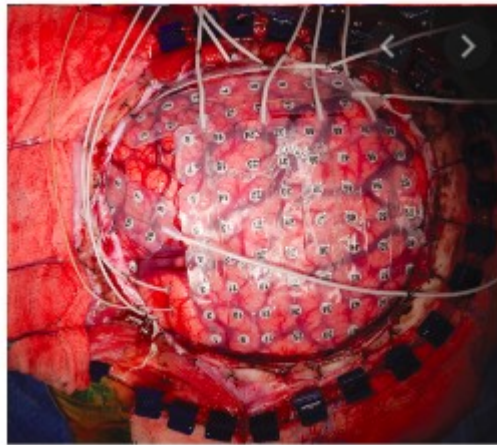


in vitro

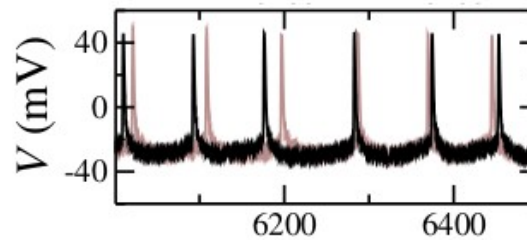
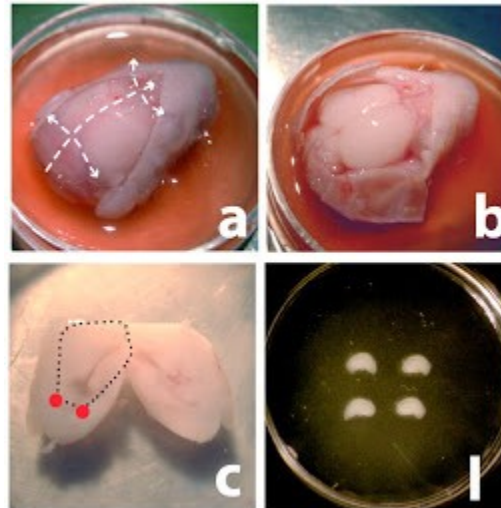


How the brain interprets reality?

in vivo



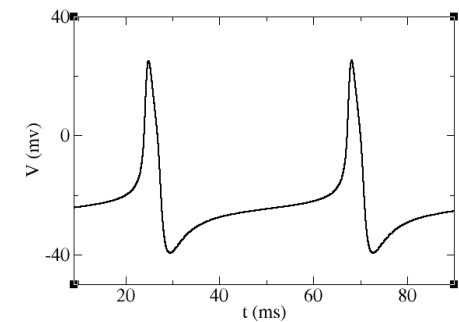
in vitro



in silico

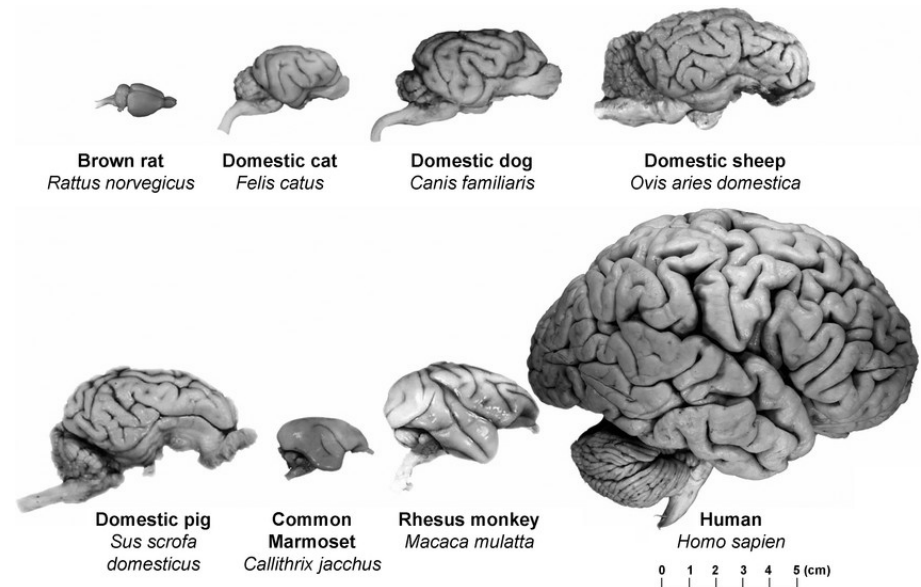
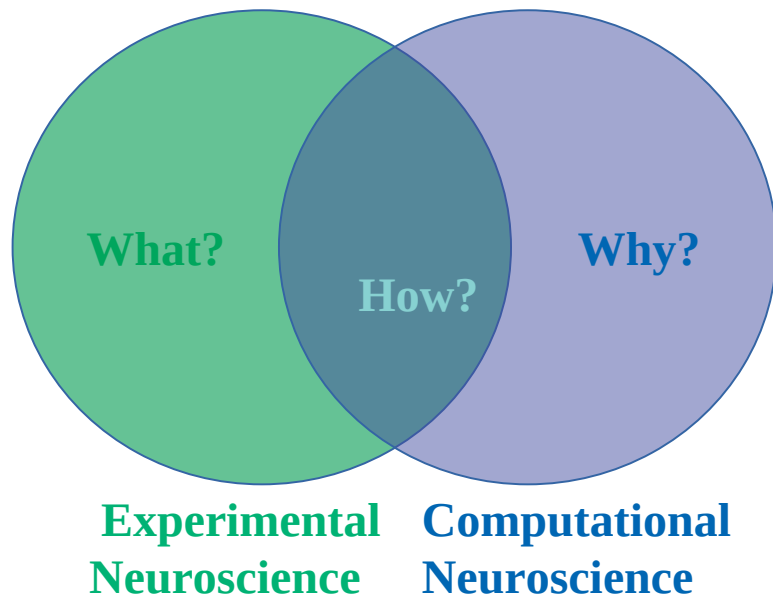


$$C_m \frac{dV}{dt} = \overline{G}_{Na} m^3 h (E_{Na} - V) + \overline{G}_K n^4 (E_K - V) + G_m (V_{rest} - V) + I + \sum I_{syn}$$



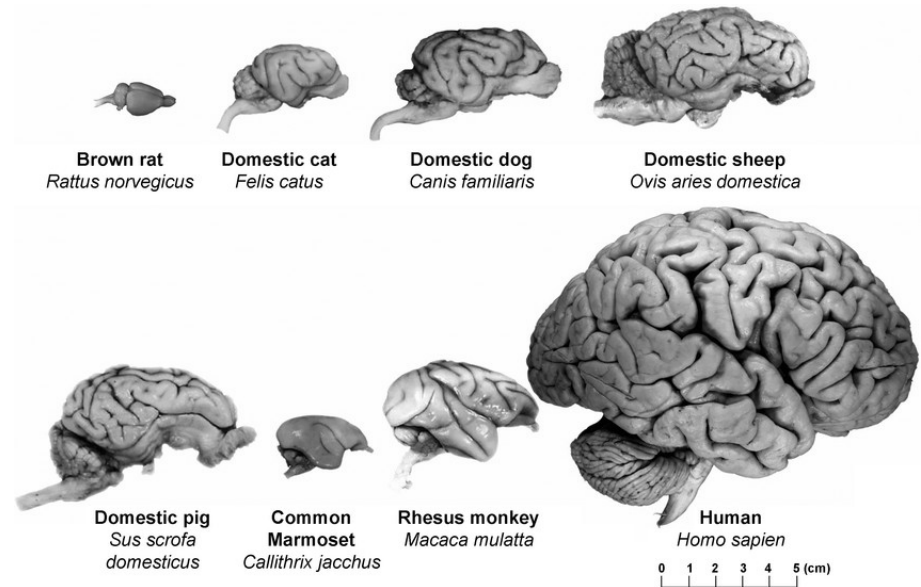
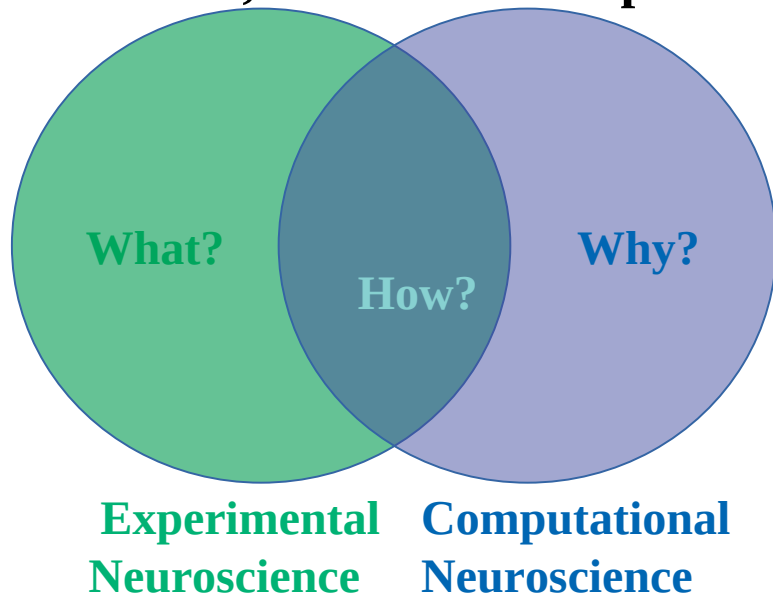
How the brain interprets reality?

One could say that the main goal of computational neuroscience is to explain how the electrical and chemical signals in the cells and interconnected networks of cells in the brain are used to represent and process the sensory stimuli around us, and how we respond to these stimuli.



How the brain interprets reality?

One could say that the main goal of computational neuroscience is to explain how the electrical and chemical signals in the cells and interconnected networks of cells in the brain are used to represent and process the sensory stimuli around us, and how we respond to these stimuli.



Can we build a machine that interpret reality?

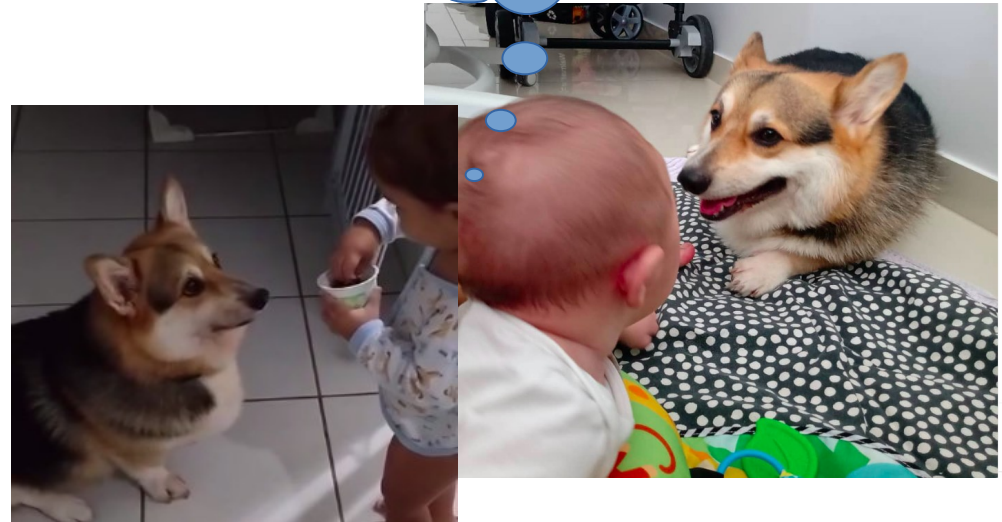
Can we build a machine that interpret reality?



Can we build a machine that interpret reality?

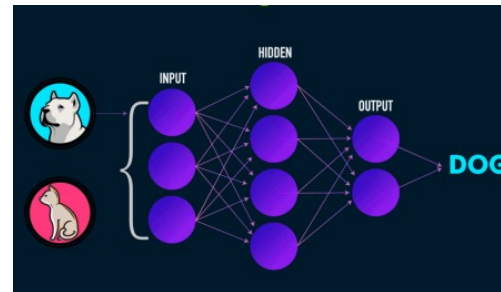


Actually, I built two machines that interpret reality
AND classify dogs!

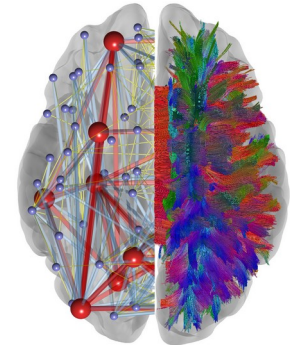


Best Biological Neural Networks I have
implemented so far

Can we use math and computers to build a machine that is similar to the brain?



Artificial Neural
Networks (ANN)



Biological Neural
Networks (BNN) Models

Can we use math and computers to build a machine that is similar to the brain?

Cognitive tasks

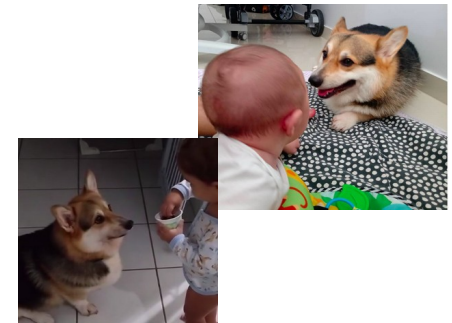
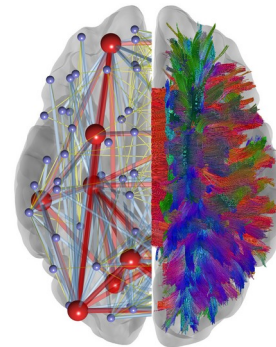
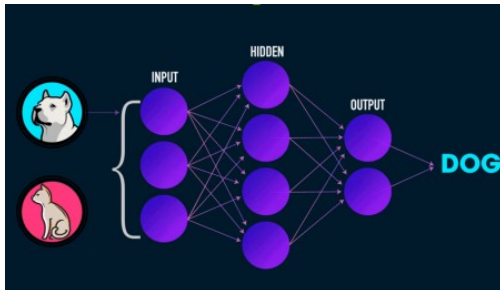
↑
Interpret reality

?

?



Classify dogs



Artificial Neural
Networks (ANN)

Biological Neural
Networks (BNN) Models

Real brains

Biological Neural Networks

Can we use math and computers to build a machine that is similar to the brain?

Cognitive tasks

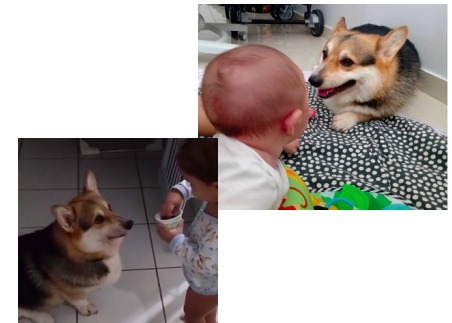
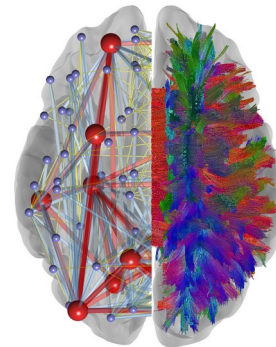
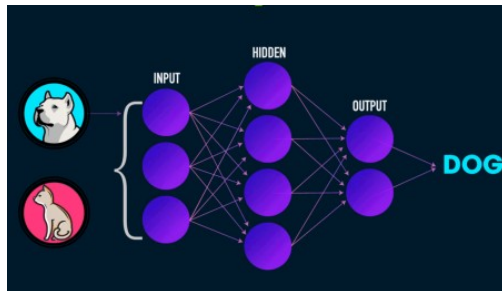
Interpret reality

What is the simplest model to reproduce a task?
What model is more efficient for a specific task?

What are the important biological ingredients?
How does the brain perform the task?



Classify dogs



Artificial Neural
Networks (ANN)

Biological Neural
Networks (BNN) Models

Real brains

Biological Neural Networks

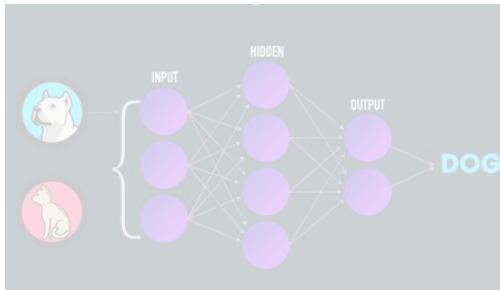
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Cognitive tasks

↑
Interpret reality

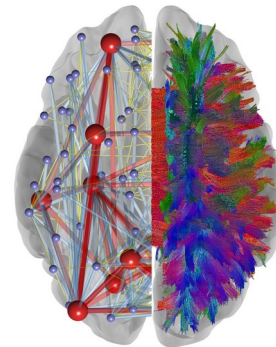
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Classify dogs

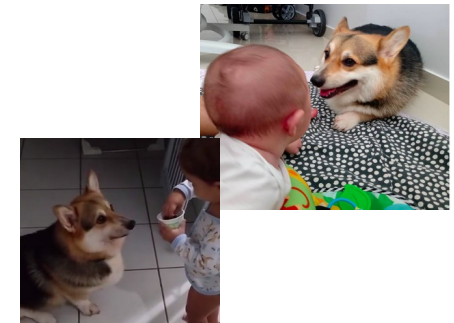


Artificial Neural
Networks (ANN)

What are the important
biological ingredients?
How does the brain
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Biological Neural
Networks (BNN) Models



Real brains

Biological Neural Networks

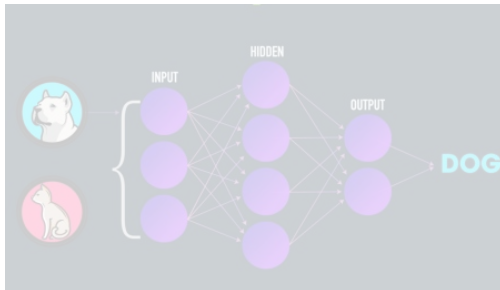
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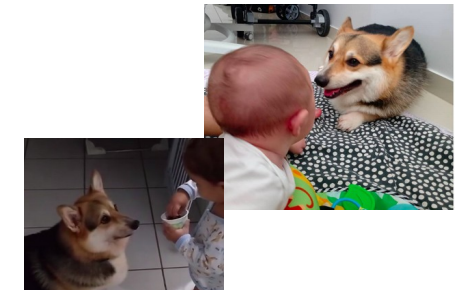
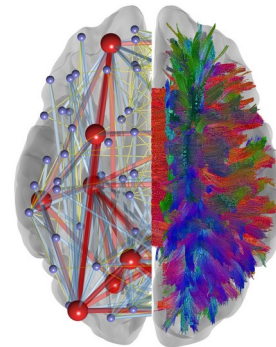
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Biological Neural
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Real brains

Biological Neural Networks

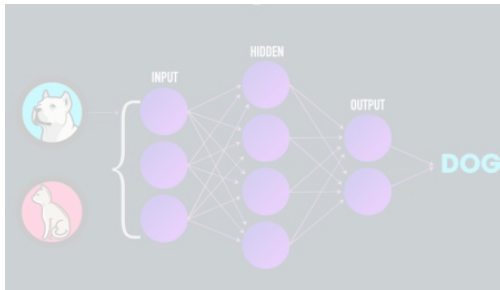
First example: The binding problem

Cognitive tasks

↑
Interpret reality

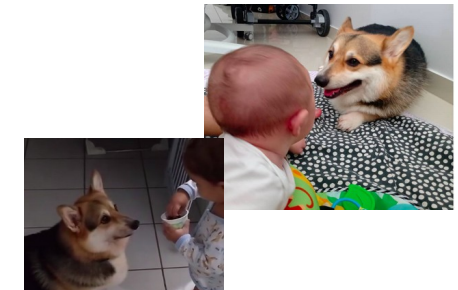
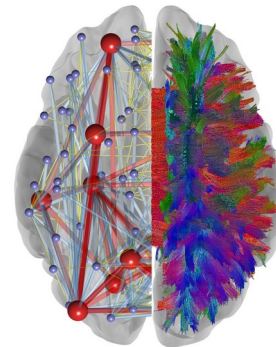
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Real brains

Biological Neural Networks

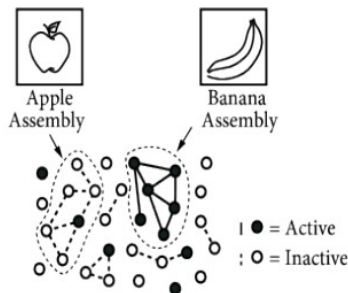
The binding problem: this is an apple!



- The brain is a good classifier but it also does other things
- How does the brain perform the task of recognize an object?
- Does neurons for color, shape, food fire together?
- What other memories are activate when we see an apple?

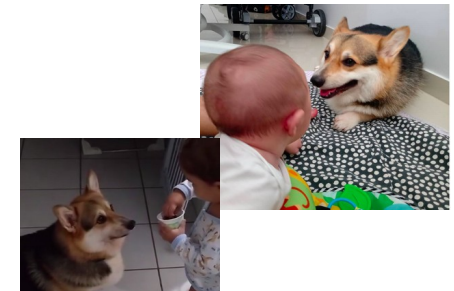
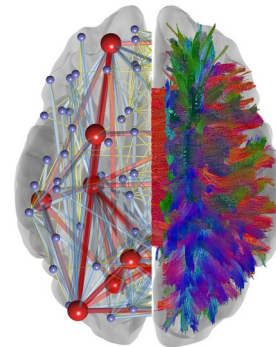
Binding by synchrony proposes that neurons representing different features of the same object fire in synchrony, enabling perceptual integration.

“Neurons that fire together wire together”



Neuronal assemblies / Hebbian rules

What are the important biological ingredients?
How does the brain perform the task?



Biological Neural
Networks (BNN) Models

Real brains

Biological Neural Networks

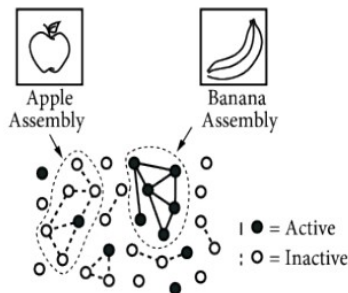
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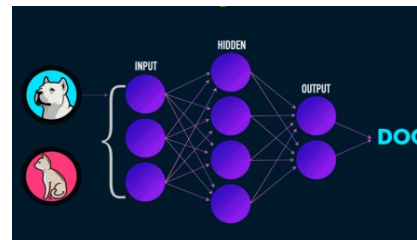
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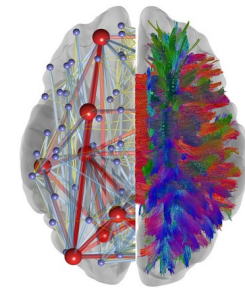


Neuronal assemblies / Hebbian rules

Can we build a machine that classifies an apple?



Artificial Neural Networks (ANN)



Biological Neural Networks (BNN)

How the brain classifies an apple?

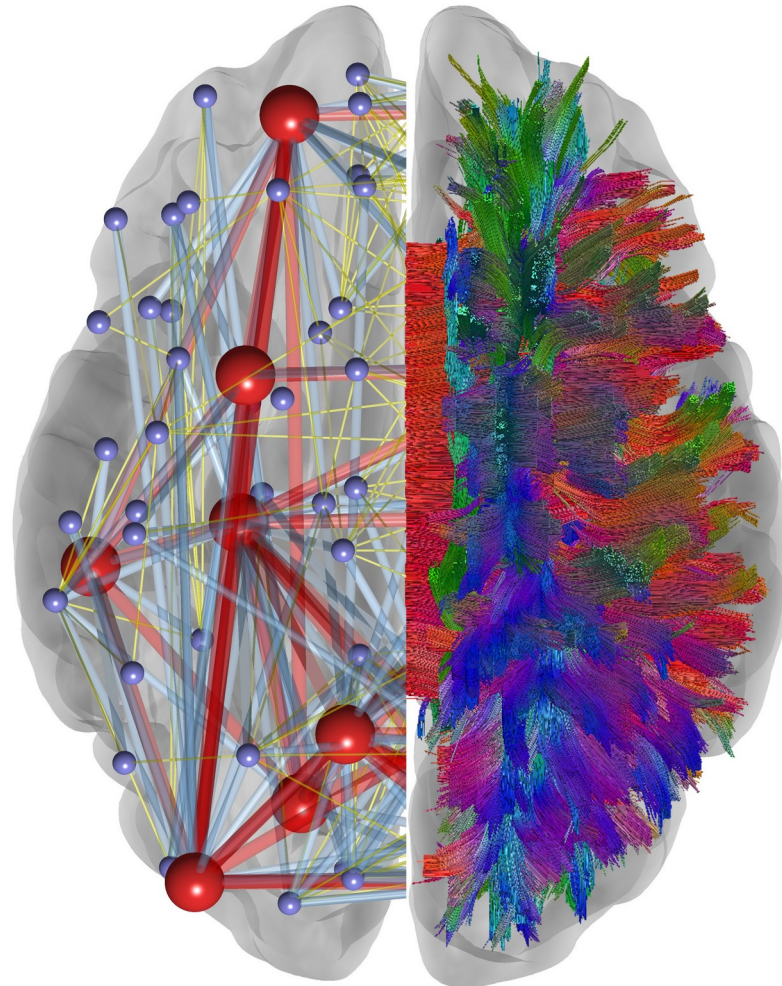
Biological Neuronal Networks: Functional x Structural Connectivity

How do neurons
communicate with
each other?

Where does the
information go?

Correlation and
causality

Changes that depend
on the task or
cognitive state
(happen very quickly)



How are neurons
connected to each other?

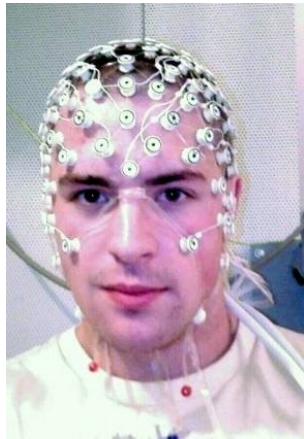
What is the network
topology?

Real synapses (only
measured post-mortem)

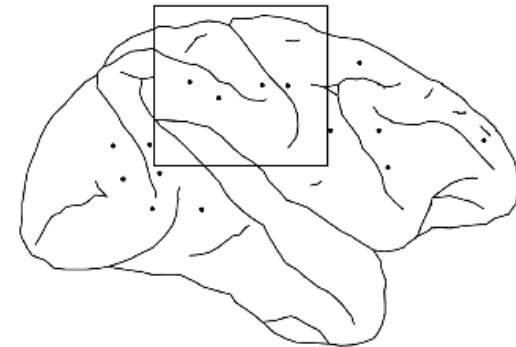
Changes depend on
synaptic plasticity
(can be slow)



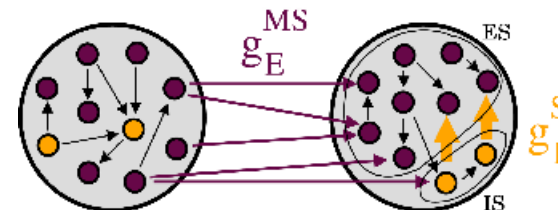
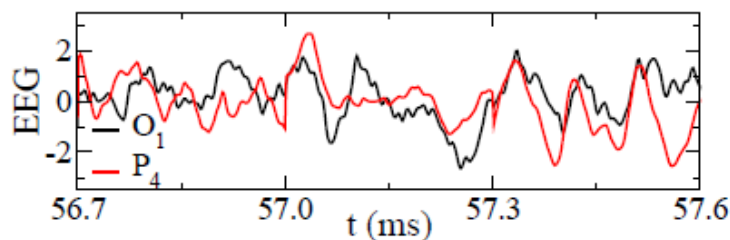
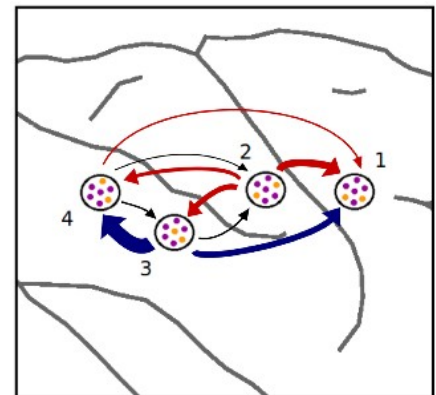
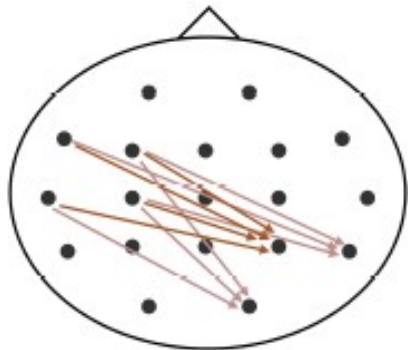
Functional x Structural Connectivity



Determine functional connectivity from electrical signals by calculating correlation, causality, phase, entropy complexity.

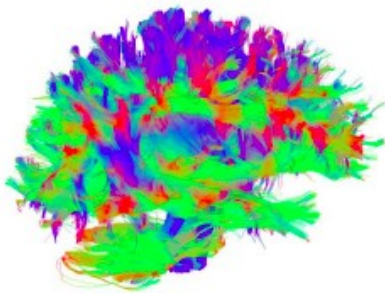


Develop plausible biological neural networks models (with known structural connectivity) that exhibit functional connectivity similar to experimental data.

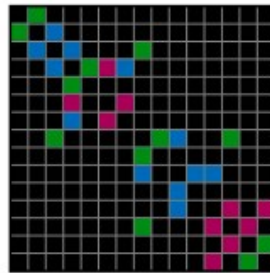


Biological Neural Networks: structural connectivity

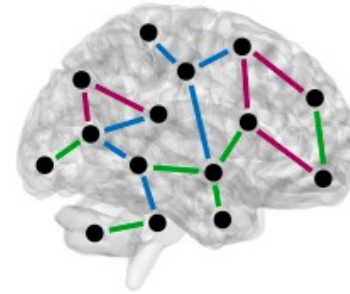
a Measurement



Example: White matter tracts (via DTI)

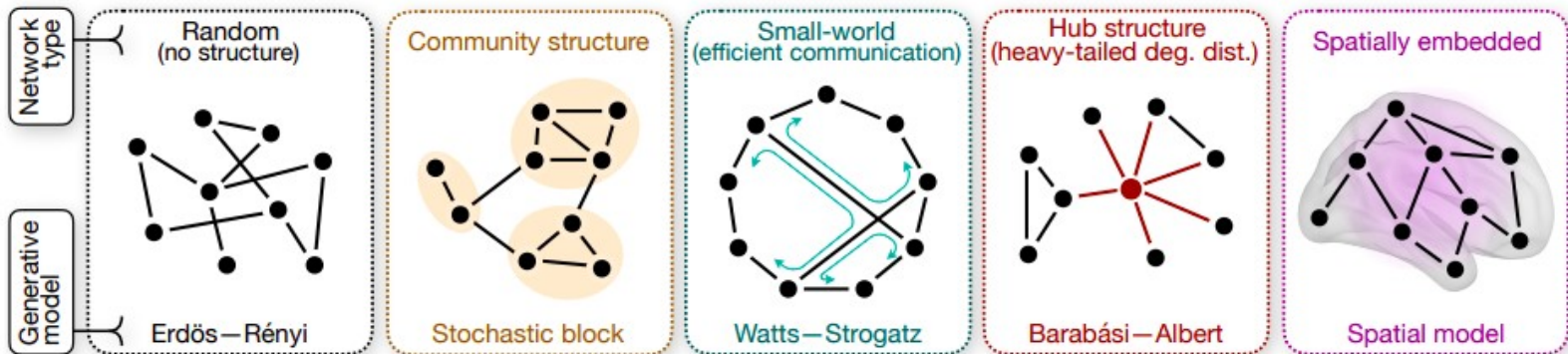


Adjacency matrix

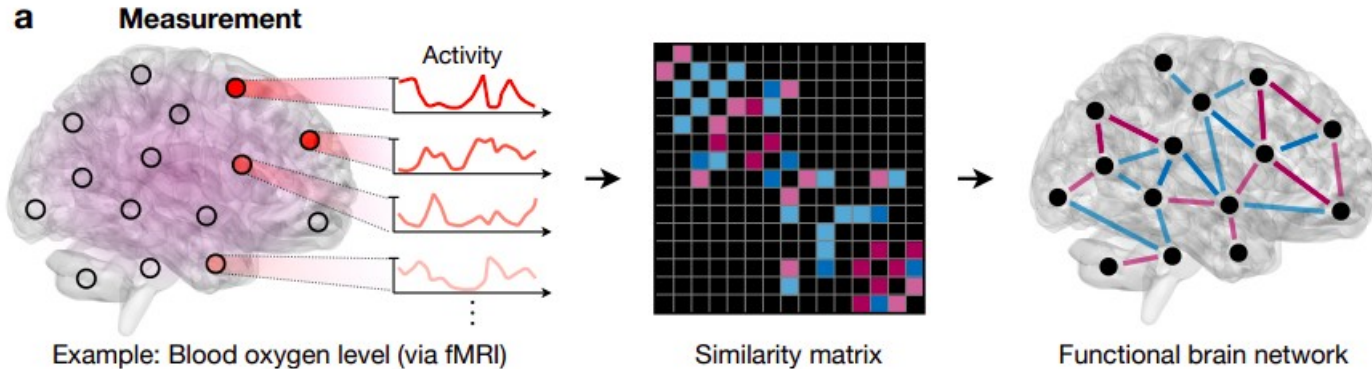


Structural brain network

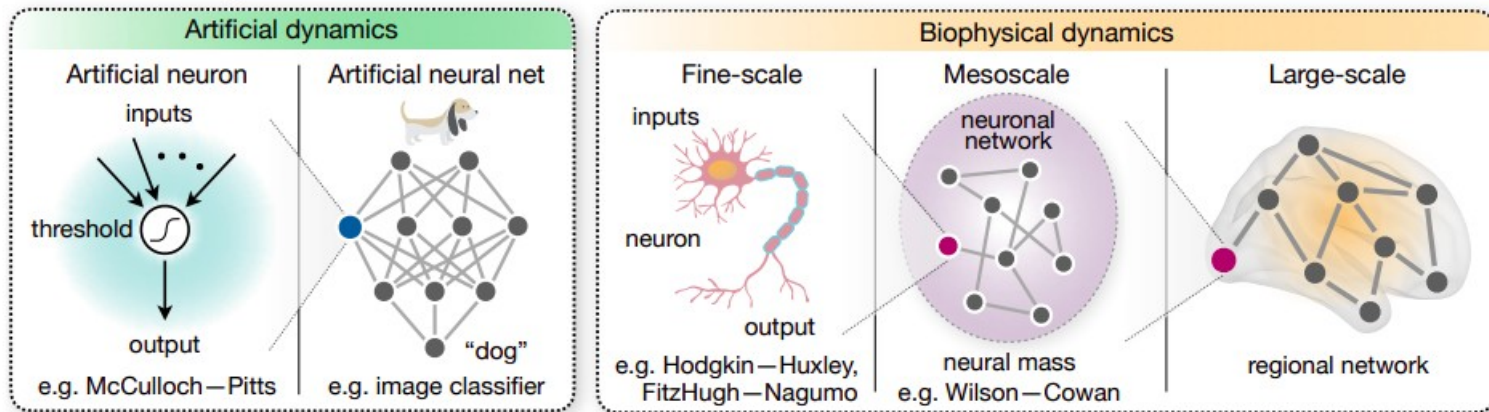
b Modeling



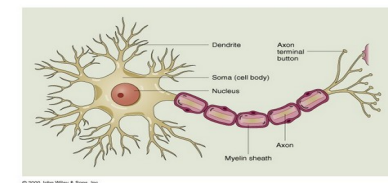
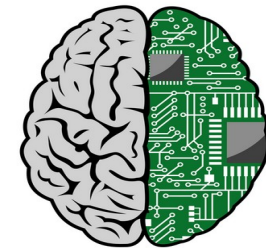
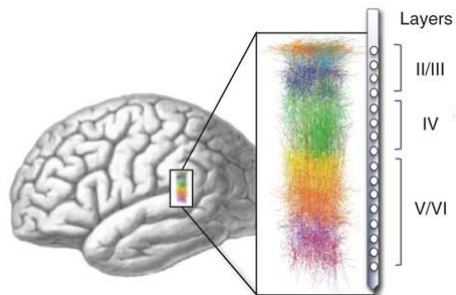
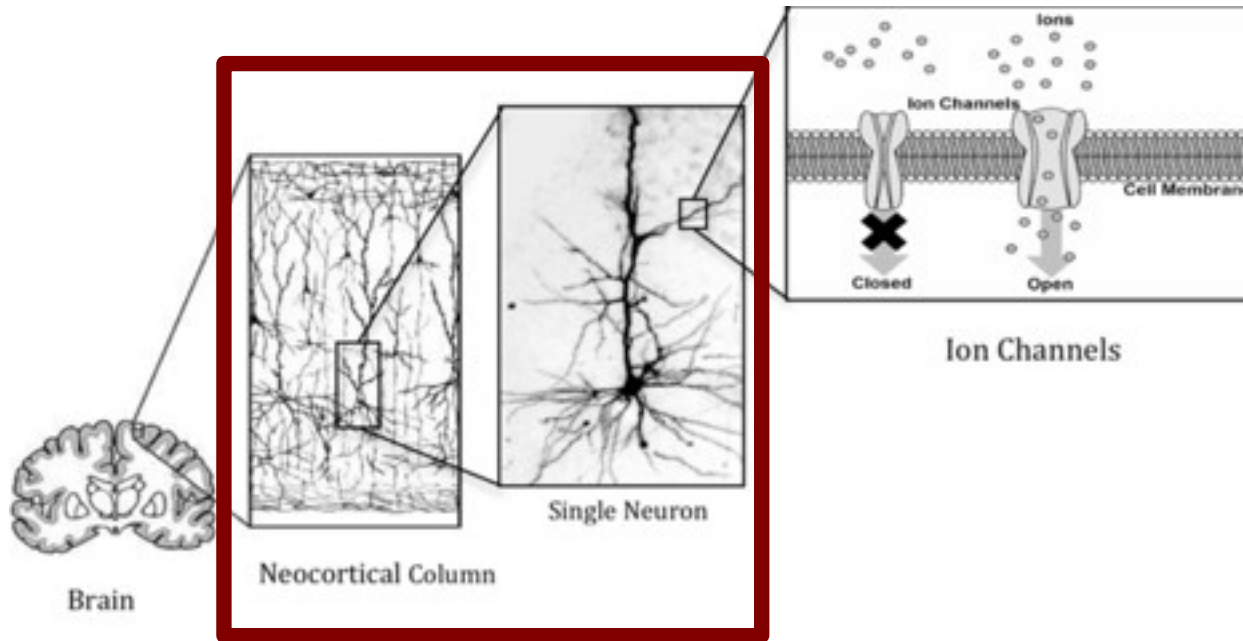
Biological Neural Networks: functional connectivity



b Modeling



Multiple scales



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Multiple scales

meters

10^0

Cognition and Nervous system

10^{-1}

Brain (1,3 Kg)

10^{-3}

**brain regions:
cortex, thalamus, hippocampus**

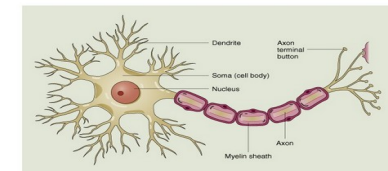
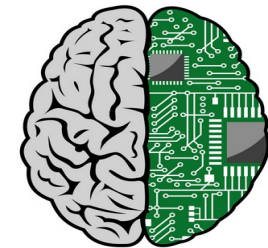
10^{-4} - 10^{-6}

celular level: neurons and synapse

10^{-9} - 10^{-10}

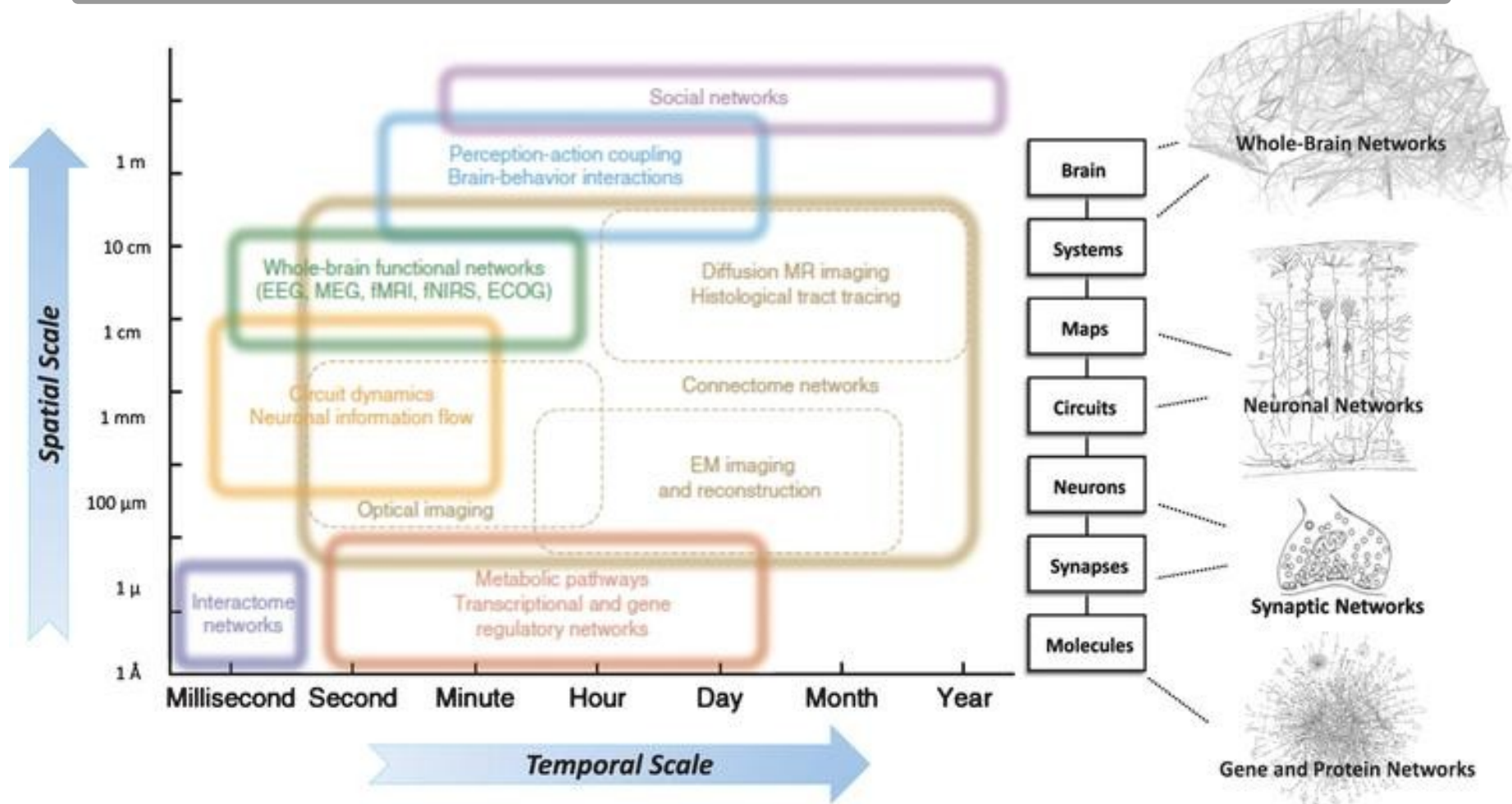
**molecular level: neutoransmitter,
biochemistry**

**We have around 10^{11} neurons;
 10^4 connections/neuron**

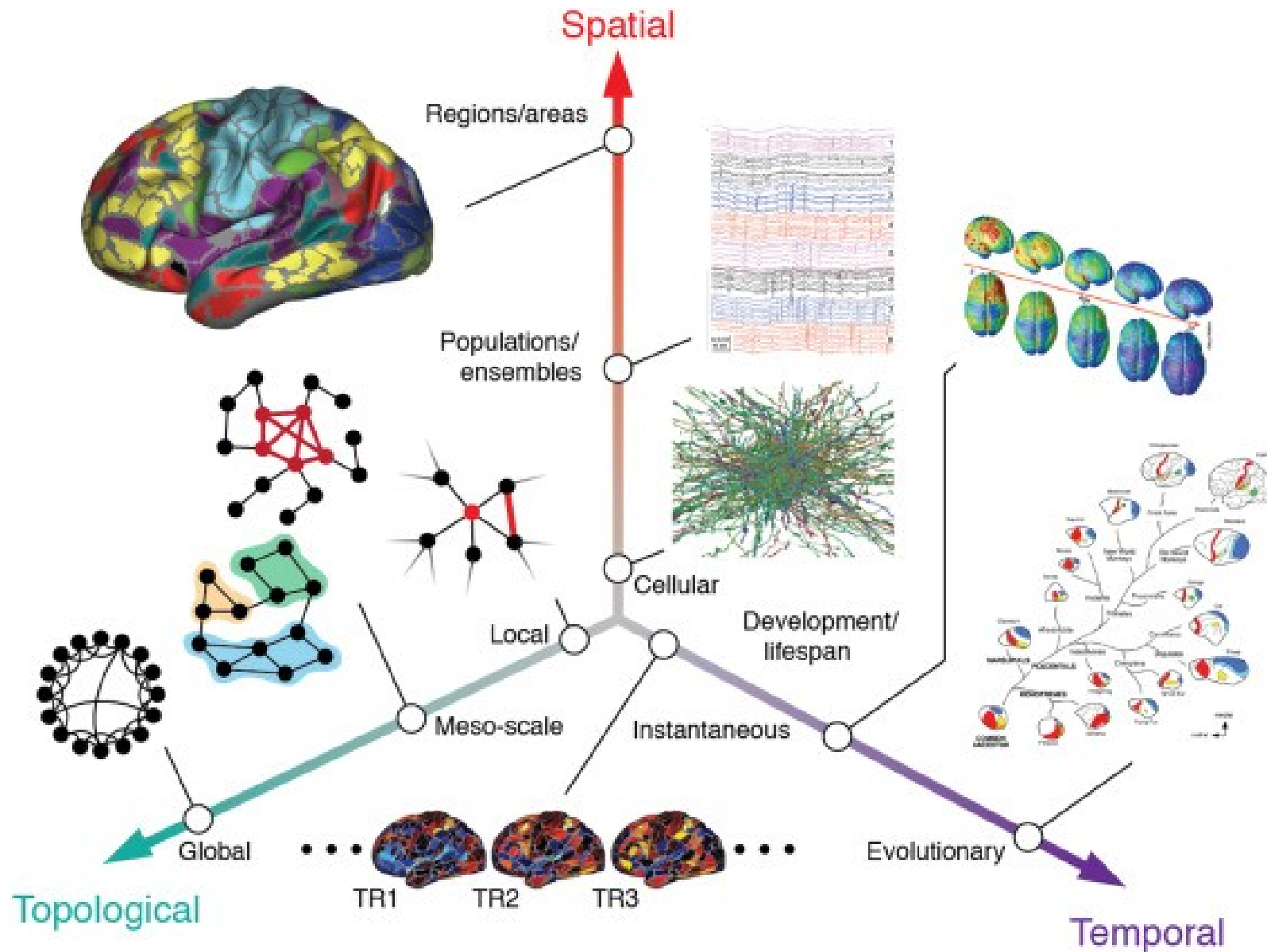


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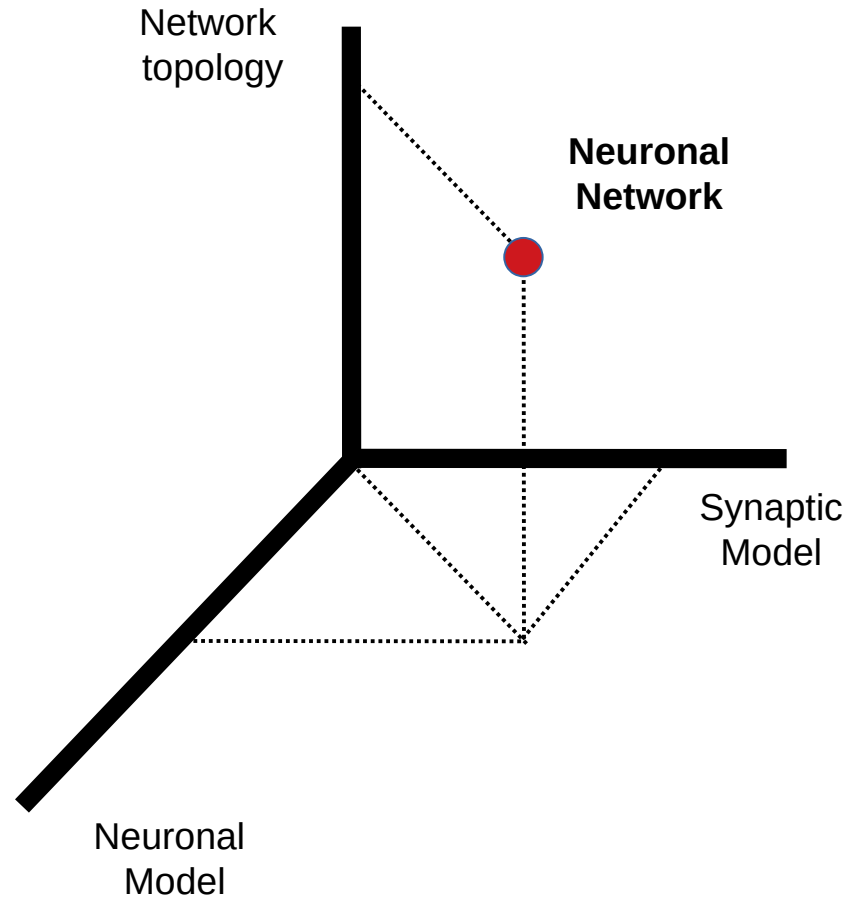
Multiple scales



Multiple scales

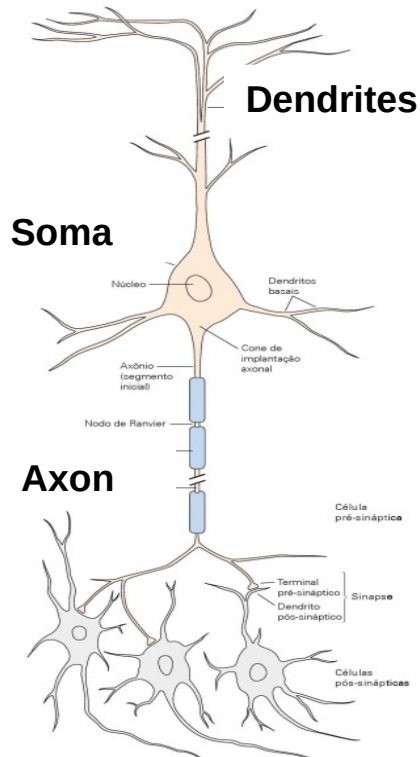
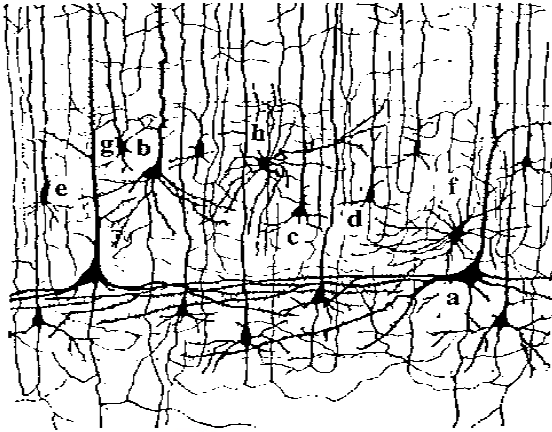


Biological Neuronal Network



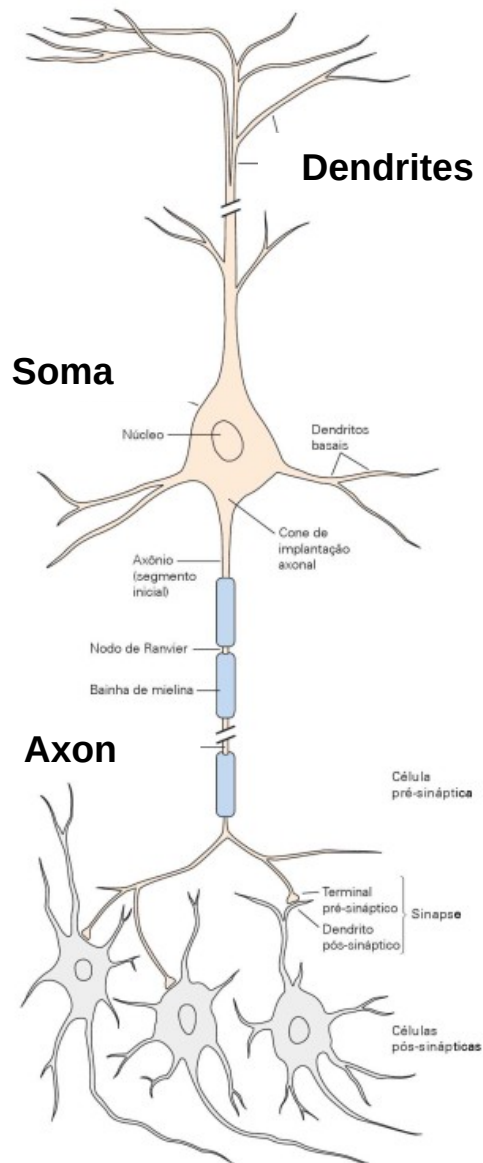
Neuronal Models

Neurons

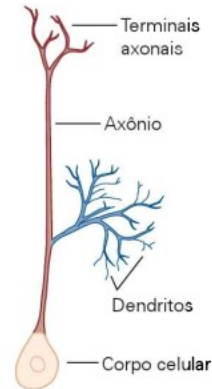


- The brain is made of isolated cells (neurons and glia), which are structurally, metabolically and functionally independent.
- Neuron doctrine (Ramon y Cajal, 1894):
The neuron is the basic functional unit of the nervous system
- Neurons are specialized for intercellular communication (synapses)
- The cortex has around 10.000 neurons and 3km of wire in a volume of 1mm³
- Neuronal structure:
 - Dendrites: receive and process inputs
 - Soma (cell body): integrates inputs
 - Axon: generates output (at the initial segment) and transmits it to target neurons
 - The connection from an axon of one neuron to a dendrite of another neuron is called a synapse

Neurons: diversity of forms and functions

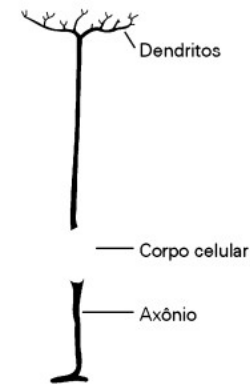


A Célula unipolar



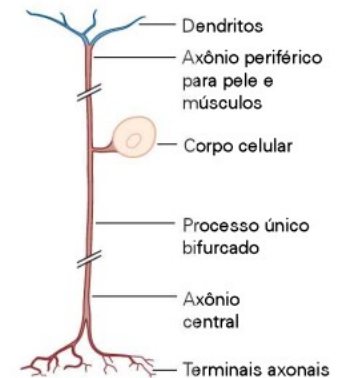
Neurônio de invertebrado

B Célula bipolar



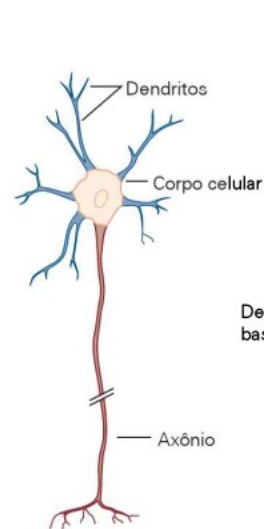
Célula bipolar da retina

C Célula pseudounipolar

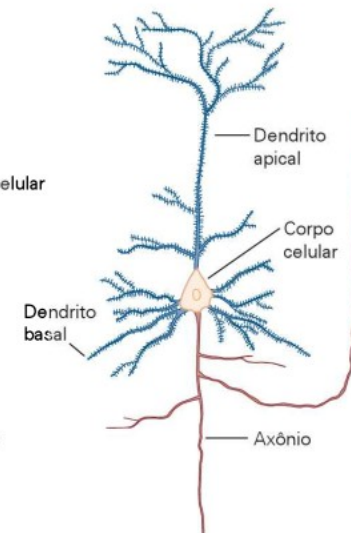


Célula ganglionar da raiz dorsal

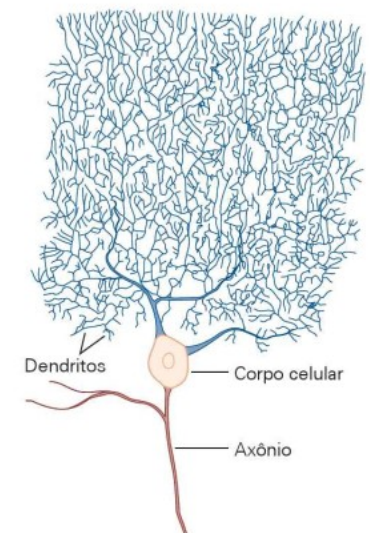
D Três tipos de células multipolares



Neurônio motor da medula espinal



Célula piramidal do hipocampo



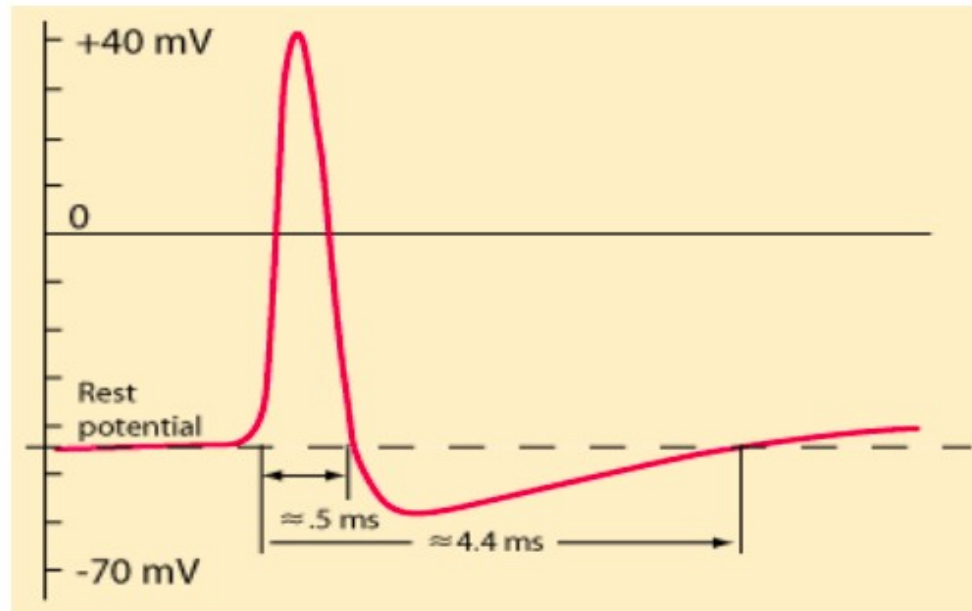
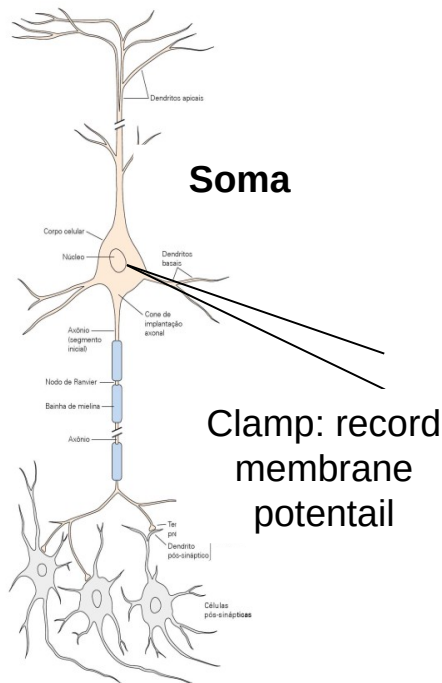
Célula de Purkinje do cerebelo

Signal: Action Potential (SPIKE)

- A rapid depolarization to positive potentials (< 0.5 ms),
- Followed by an almost equally rapid repolarization (< 0.5 ms)
- and a longer after-hyperpolarization (< 4 ms): refractory period

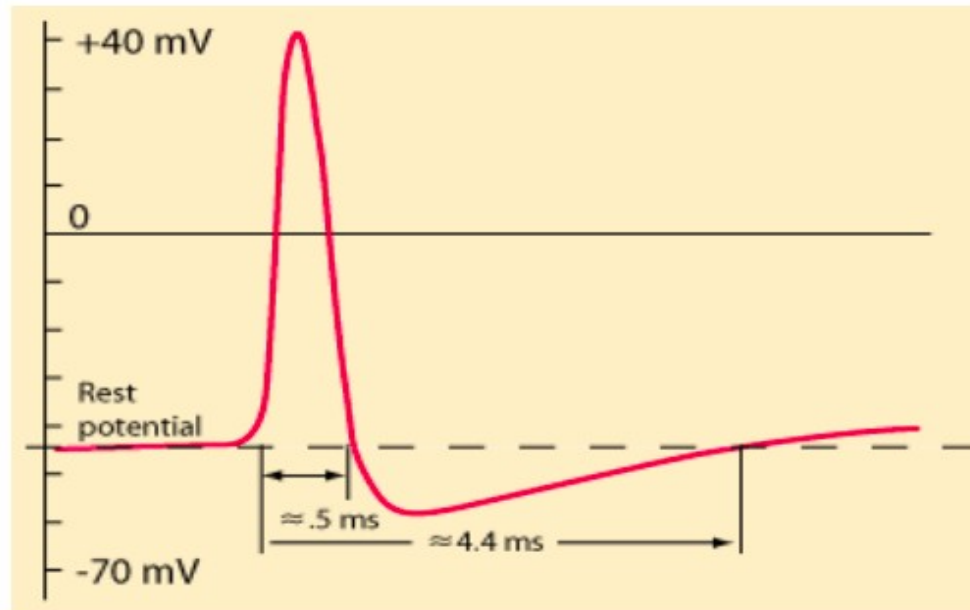
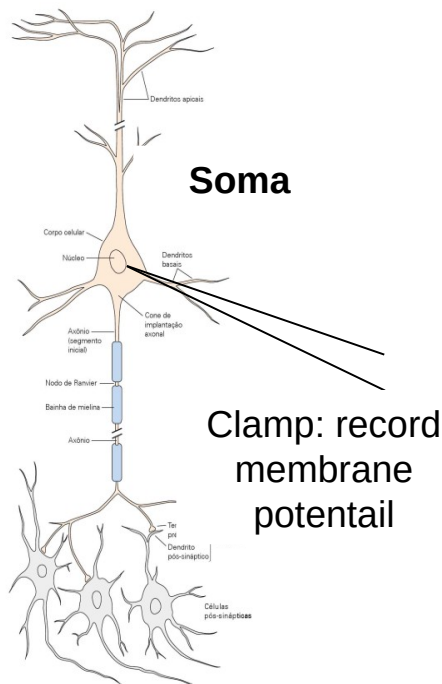
Absolute: period during which a second stimulus (no matter how strong) will not lead to a second spike. It is as if the spike threshold were infinite

Relative: period during which a second spike can be generated by a second stimulus stronger than the first.



Signal: Action Potential (SPIKE)

- Shape (width and amplitude) characteristic of each neuron
- Threshold phenomenon (all or none)
- Propagates unchanged along the axon
- Subthreshold voltage fluctuations are strongly attenuated

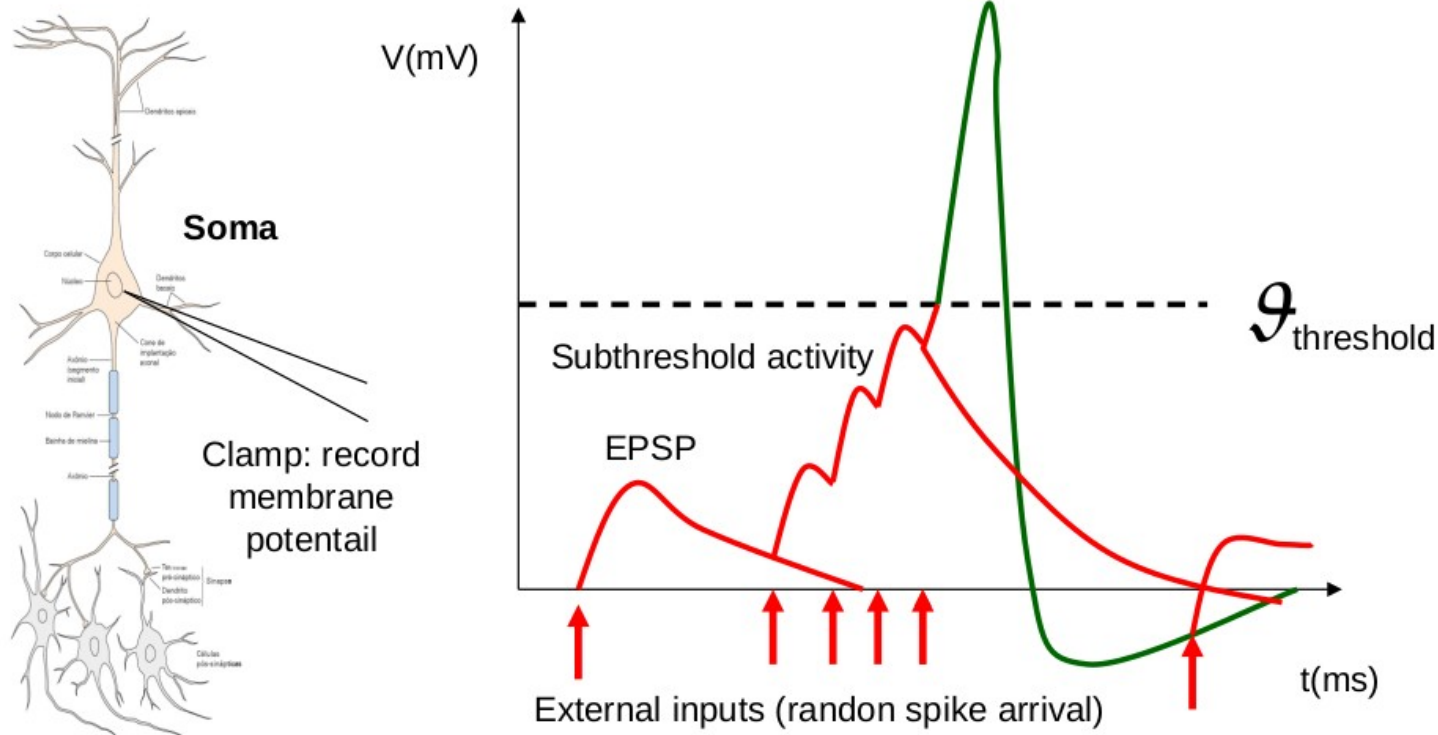


Signal: Action Potential (SPIKE)

Output signal: spikes are **rare** events triggered at threshold

Subthreshold regime: trajectory of potential shows fluctuations

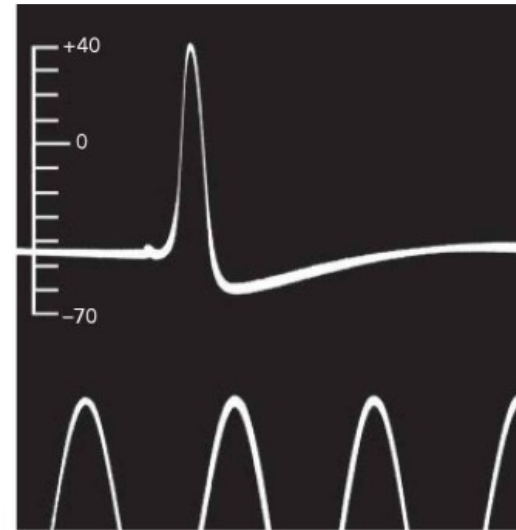
EPSP: excitatory post synaptic potential



The Hodgkin Huxley neuron model (1952)

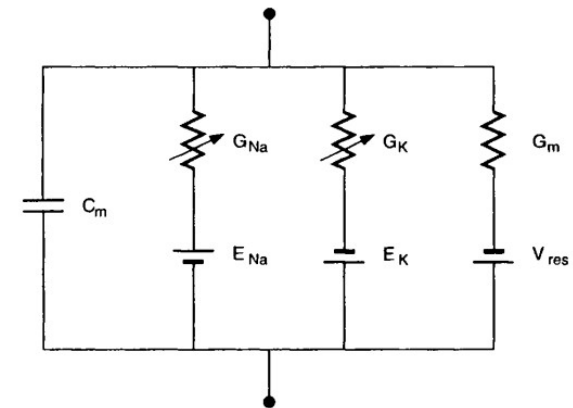
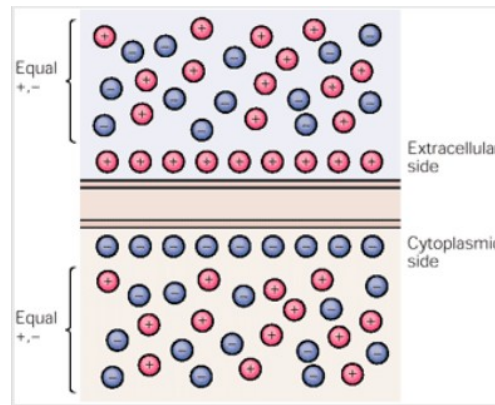
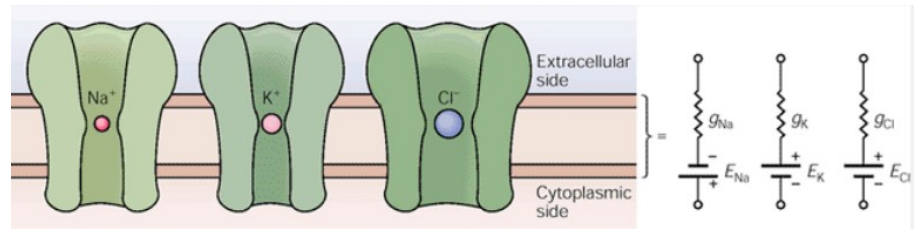
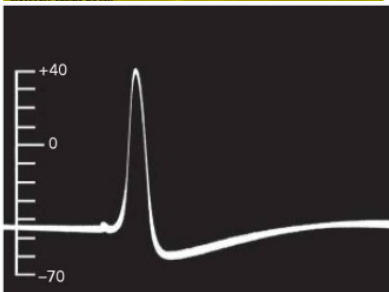


First action potential



- The neuron exhibits electrical activity
- It exchanges ions with the external environment
- There is a variation in the potential difference between the internal and external environments
- HH experimentally measure this variation

Let's think about an electrical circuit!



- The neuron exhibits electrical activity
- It exchanges ions with the external environment
- There is a variation in the potential difference between the internal and external environments
- HH experimentally measure this variation
- **They develop a mathematical equation to describe this variation (a model)**

$$V_m = V_{in} - V_{out}$$

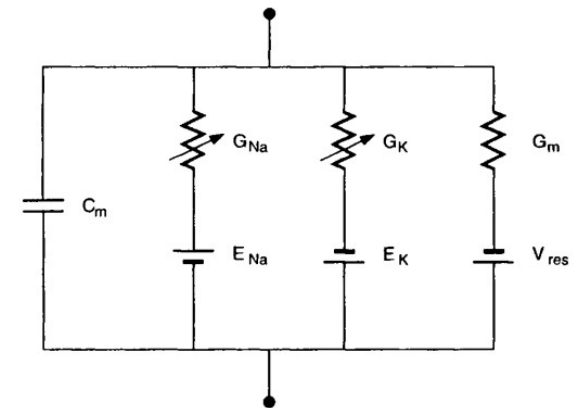
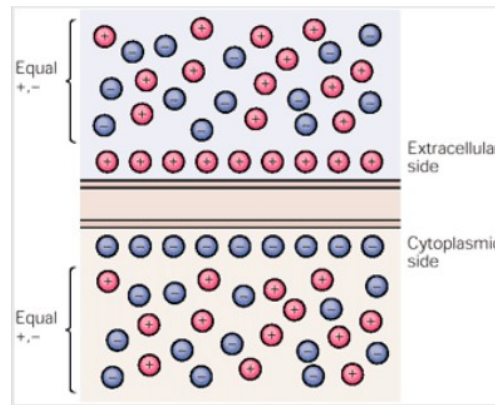
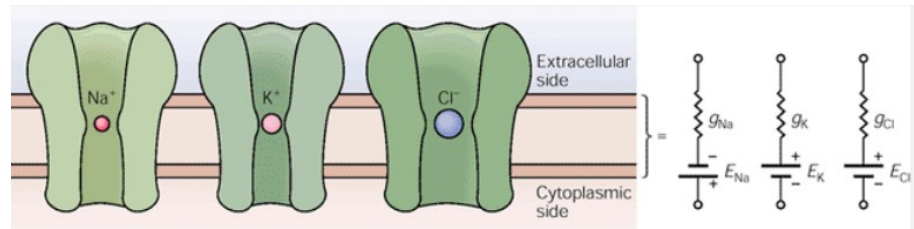
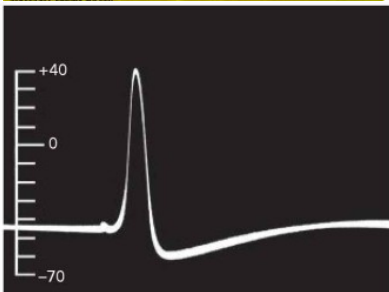
$$V = Q/C$$

$$\Delta V = \Delta Q/C$$

$$\Delta V = I_c \cdot \Delta t / C$$

$$C_m \frac{dV}{dt} = \sum I$$

Let's think about an electrical circuit!



$$C_m \frac{dV}{dt} = \bar{G}_{Na} m^3 h (E_{Na} - V) + \bar{G}_K n^4 (E_K - V) + G_m (V_{rest} - V) + I + \sum I_{syn}$$

$$\frac{dx}{dt} = \alpha_x(V)(1 - x) - \beta_x(V)x$$

$$V_m = V_{in} - V_{out}$$

$$V = RI$$

$$V = Q/C$$

$$R = 1/g$$

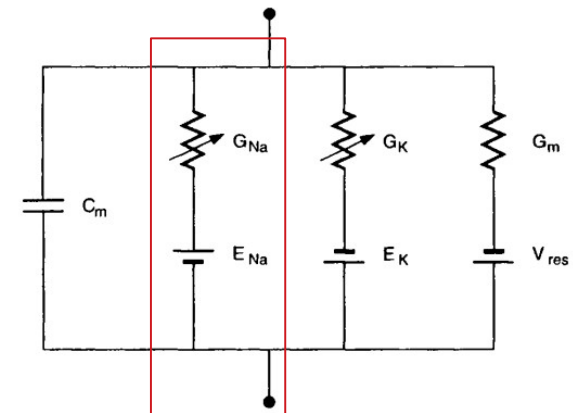
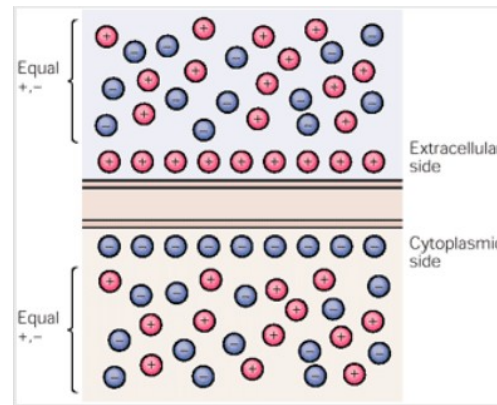
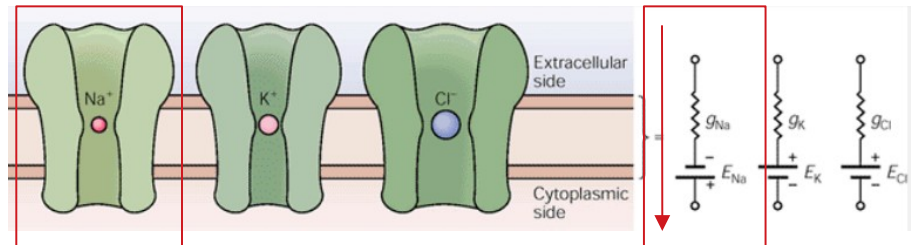
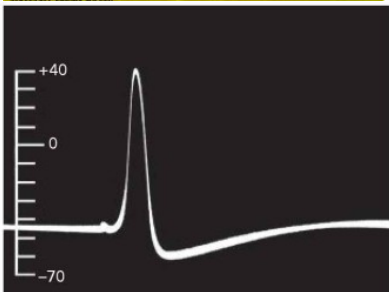
$$\Delta V = \Delta Q/C$$

$$I = gV$$

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$$C_m \frac{dV}{dt} = \sum I$$

Let's think about an electrical circuit!



$$C_m \frac{dV}{dt} = \overline{G}_{Na} m^3 h (E_{Na} - V) + \overline{G}_K n^4 (E_K - V) + G_m (V_{rest} - V) + I + \sum I_{syn}$$

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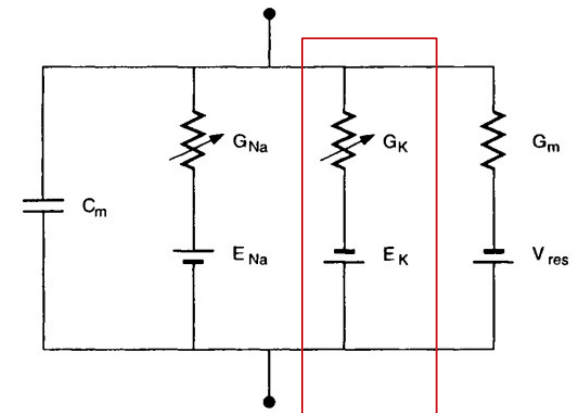
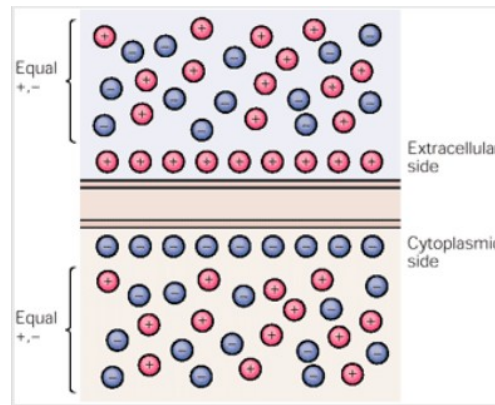
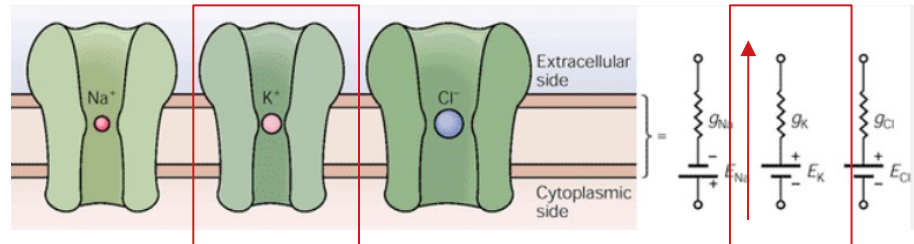
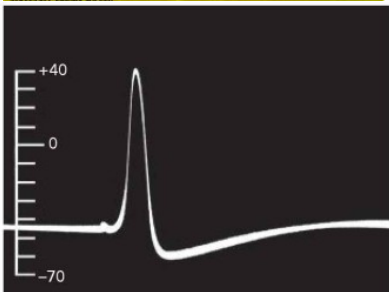
$$C_m \frac{dV}{dt} = \sum I$$

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$$C_m \frac{dV}{dt} = \overline{G}_{Na} m^3 h (E_{Na} - V) + \overline{G}_K n^4 (E_K - V) + G_m (V_{rest} - V) + I + \sum I_{syn}$$

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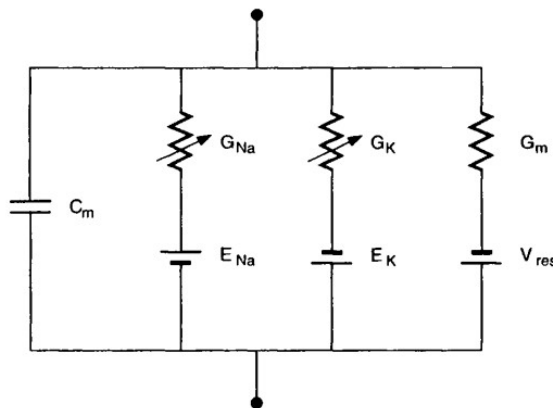
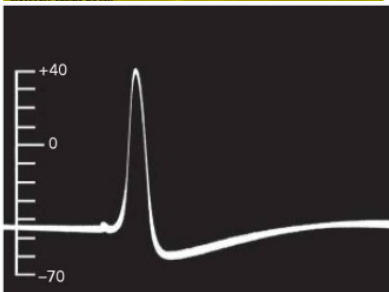
$$C_m \frac{dV}{dt} = \sum I$$

$$V = RI$$

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$$I = gV$$

The Hodgkin Huxley neuron model (1952)



$$C_m \frac{dV}{dt} = \bar{G}_{Na} m^3 h (E_{Na} - V) + \bar{G}_K n^4 (E_K - V) + G_m (V_{rest} - V) + I + \sum I_{syn}$$

$$\frac{dx}{dt} = \alpha_x(V)(1 - x) - \beta_x(V)x$$

$$n \xrightleftharpoons[\alpha_n]{\beta_n} 1 - n$$

G_i : maximal conductance

- i : fraction of G_i attained at a given moment

- E_i : Nernst potential of ion i .

- $x=m,n,h$

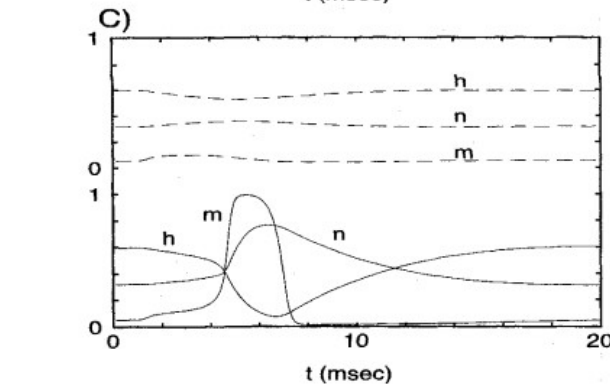
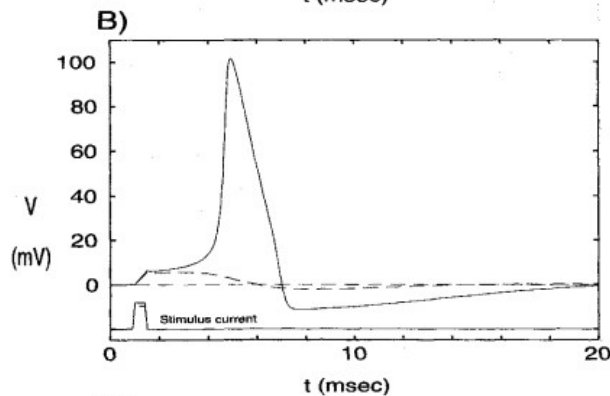
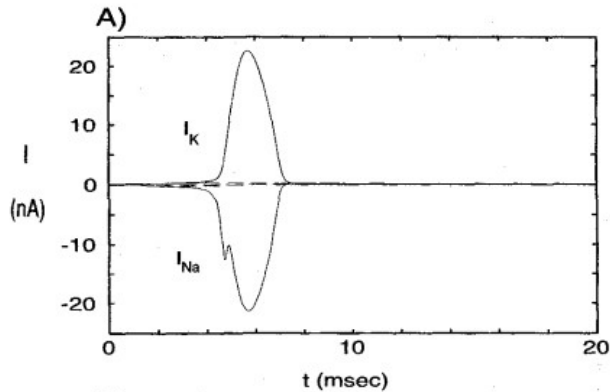
- m and n : activation variables

(grows from 0 to 1 with increase of V)

- h : inactivation variable

(decreases from 1 to 0 with increase of V)

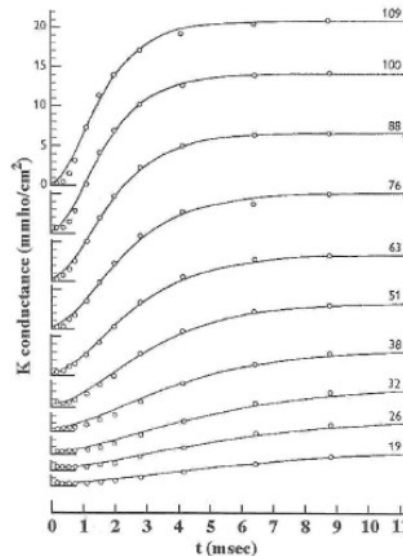
The Hodgkin Huxley neuron model (1952)



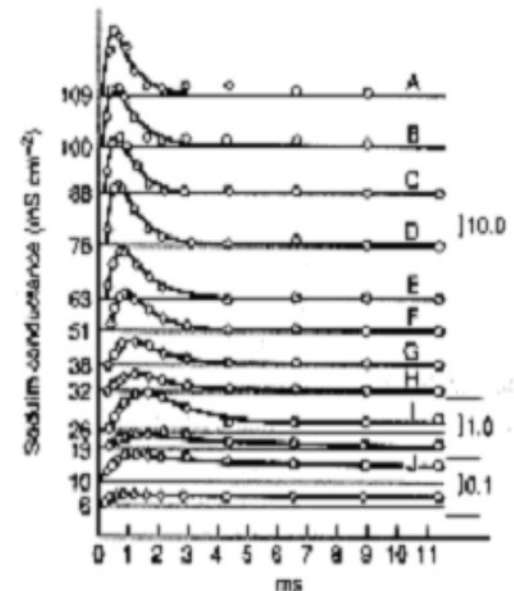
$$C_m \frac{dV}{dt} = \bar{G}_{Na} m^3 h (E_{Na} - V) + \bar{G}_K n^4 (E_K - V) + G_m (V_{rest} - V) + I + \sum I_{syn}$$

$$\frac{dx}{dt} = \alpha_x(V)(1 - x) - \beta_x(V)x$$

$$n \xrightleftharpoons[\alpha_n]{\beta_n} 1 - n$$

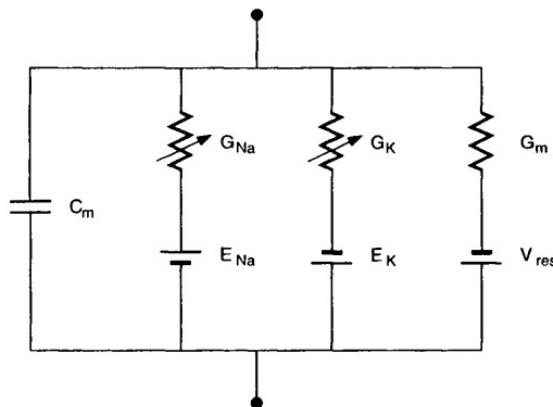
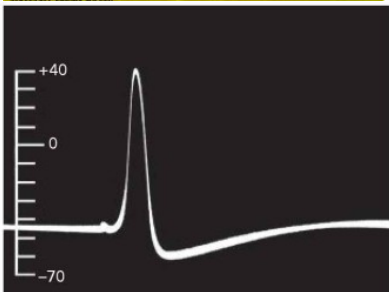


K⁺ conductance: persistent
Do not inactivate



Na⁺ conductance: transient
Inactivate

The Hodgkin Huxley neuron model (1952)



$$C_m \frac{dV}{dt} = \bar{G}_{Na} m^3 h (E_{Na} - V) + \bar{G}_K n^4 (E_K - V) + G_m (V_{rest} - V) + I + \sum I_{syn}$$

$$\frac{dx}{dt} = \alpha_x(V)(1 - x) - \beta_x(V)x$$

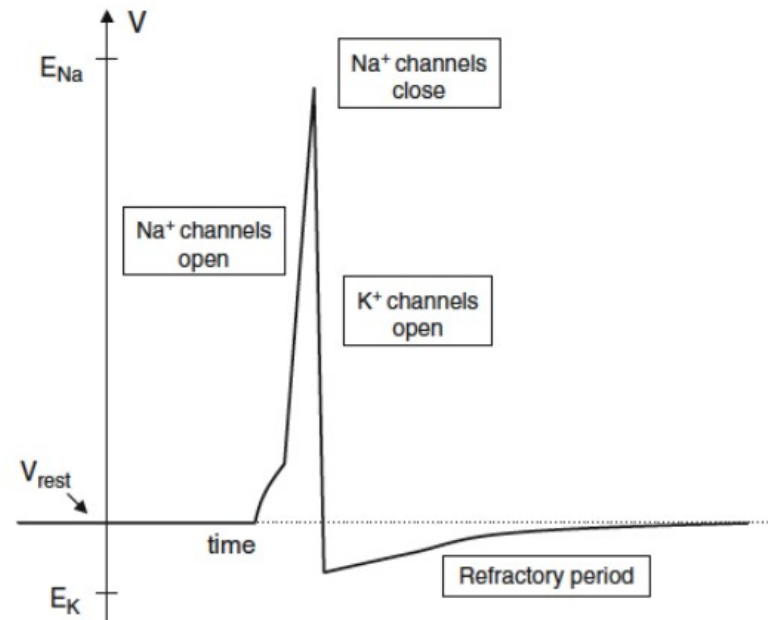
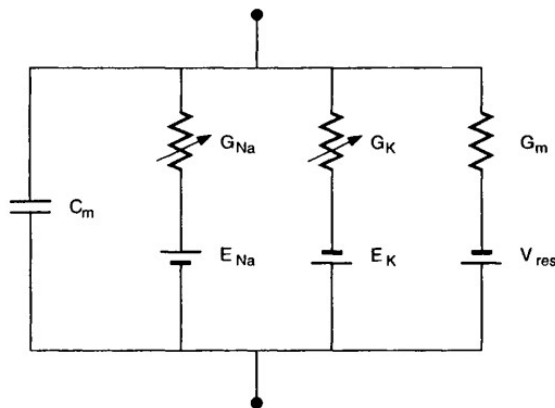
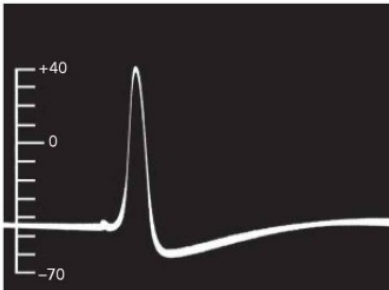


Fig. 1.7 The action potential. During the upstroke, Na^+ channels open and the membrane potential approaches the Na^+ Nernst potential. During the downstroke, Na^+ channels are closed, K^+ channels are open, and the membrane potential approaches the K^+ Nernst potential

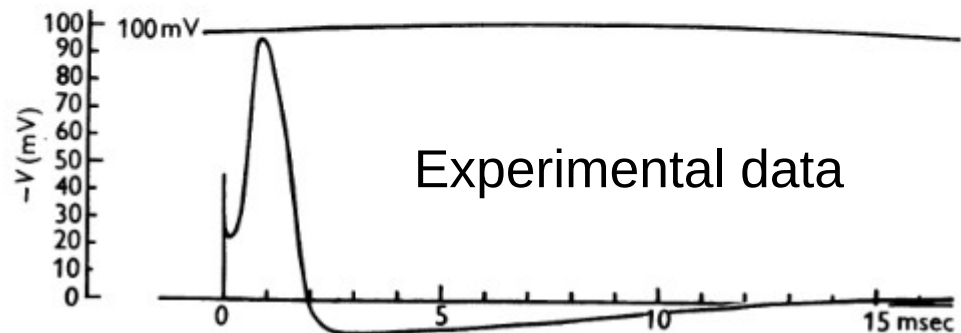
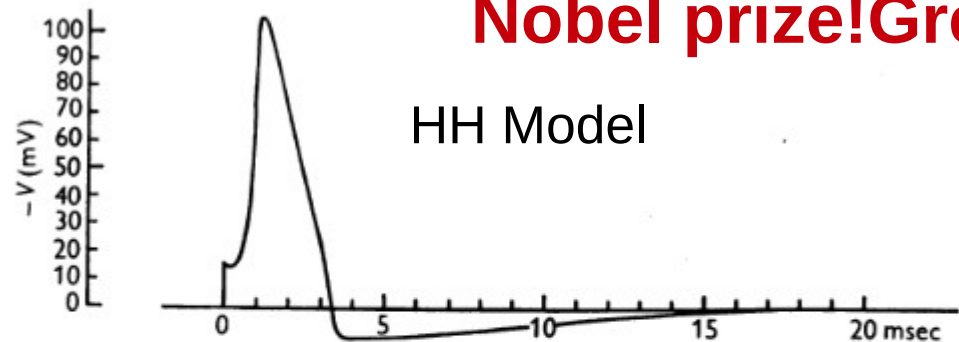
The Hodgkin Huxley neuron model (1952)



$$C_m \frac{dV}{dt} = \bar{G}_{Na} m^3 h (E_{Na} - V) + \bar{G}_K n^4 (E_K - V) + G_m (V_{rest} - V) + I + \sum I_{syn}$$

$$\frac{dx}{dt} = \alpha_x(V)(1 - x) - \beta_x(V)x$$

Nobel prize! Great!



Scientific method to develop and test models

The neuron exhibits electrical activity

It exchanges ions with the external environment

There is a variation in the potential difference between the internal and external environments

We experimentally measure this variation

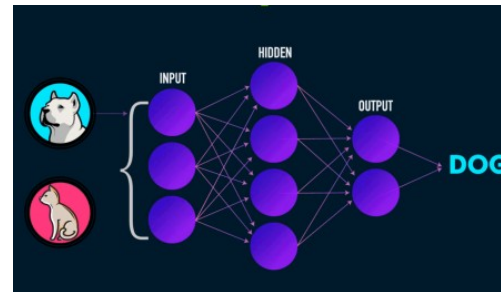
We develop a mathematical equation to describe this variation (a model)

We perform a computer simulation to reproduce the variation of the potential over time

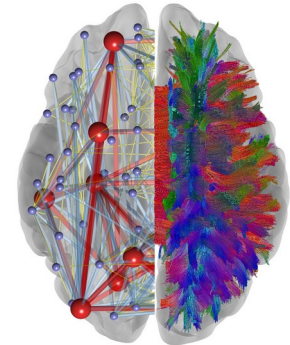
We make new predictions that can be tested experimentally



Can we build a machine that is similar to the brain?

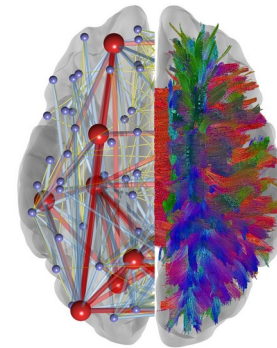
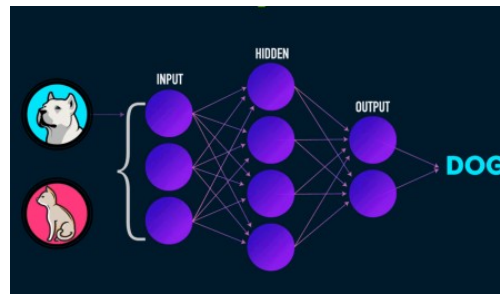


Artificial Neural
Networks (ANN)



Biological Neural
Networks (BNN) Models

Neuronal Models



Artificial Neural
Networks (ANN)

Biological Neural
Networks (BNN) Models

Simplified Models
1D or 2D Models
Firing rate

HH Model

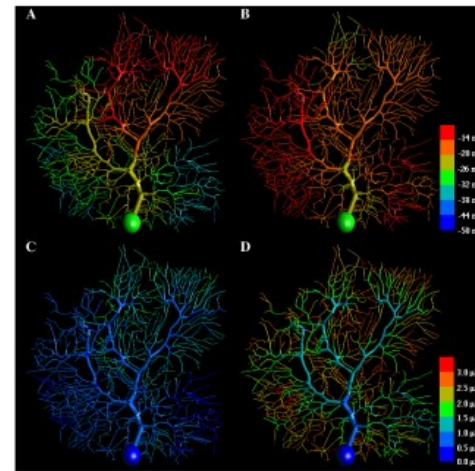
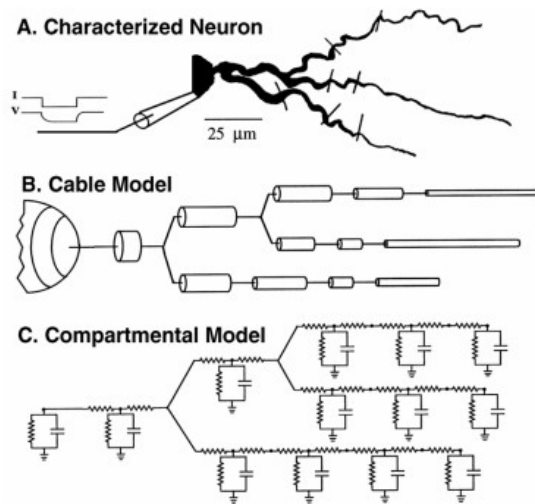
More biological ingredients
Modified HH Models
Compartmental Models

Modified Hodgkin-Huxley Models

$$C_m \frac{dV}{dt} = -g_{\text{leak}}(V - E_{\text{leak}}) - I_{\text{Na}} - I_{\text{Kd}} - I_{\text{M}} - I_{\text{T}} - I_{\text{L}}, \quad (1)$$

- Fast Na (I_{Na}): transient; action potential generation
- Slow Na (I_{NaP}): slowly inactivating (persistent); keeps neurons depolarized (facilitating bursts)
- Delayed rectifier K (I_{K}): persistent; action potential termination
- A current (I_{A}): transient K current; creates spike delays
- M current (I_{M}): slow persistent K current; causes spike rate adaptation
- Inward rectifier (I_{Kir}): activated by hyperpolarization; keeps resting potential near E_{K}
- Ca dependent K channels (I_{C} or I_{AHP}): persistent; depend on $[\text{Ca}^{2+}]$; responsible for afterhyperpolarizations (AHPs), spike frequency adaptation and bursting
- Calcium channels (I_{CaT} and I_{CaL}): transient and persistent; responsible for bursts and subthreshold voltage oscillations
- h current (I_{h}): slow; activated by hyperpolarization; mixture of Na and K ions; rhythmic firing and slow repolarization after hyperpolarizing

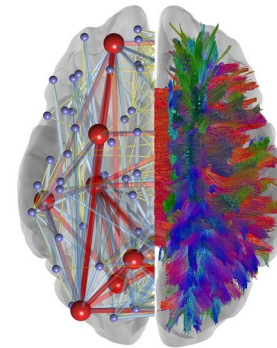
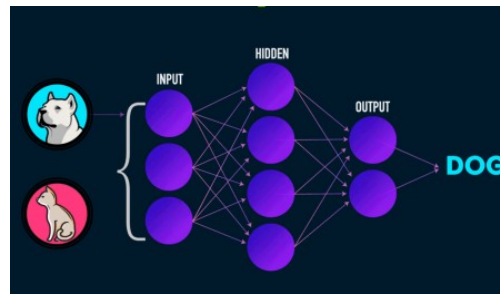
Detailed compartmental model



Model of a cerebellar Purkinje cell (De Schutter and Bower, 1994): 4550 compartments.

- Conductance-based (HH) and compartmental models are useful to study single neurons:
 - Propagation of membrane potentials along the dendritic tree
 - Pharmacological manipulations
 - Functional properties of single neurons
 - But to model neural networks it is necessary to estimate parameters for all neurons
 - Simplified models can be too simple to be used as biophysical neuron models.
- But are useful for the construction of large-scale network models and study:
- Spiking activity patterns in networks
 - Functional properties of networks or network parts

Neuronal Models



Artificial Neural
Networks (ANN)

Biological Neural
Networks (BNN) Models

Simplified Models
1D or 2D Models
Firing rate

HH Model

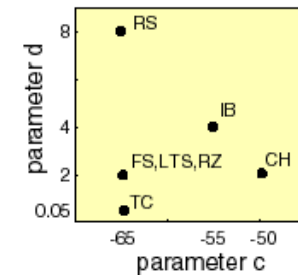
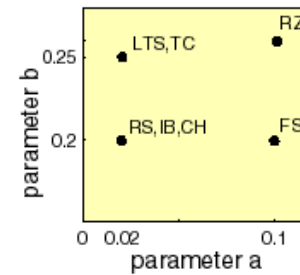
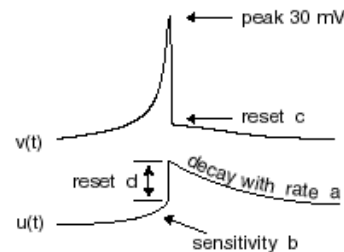
More biological ingredients
Modified HH Models
Compartmental Models

The Izhikevich Model

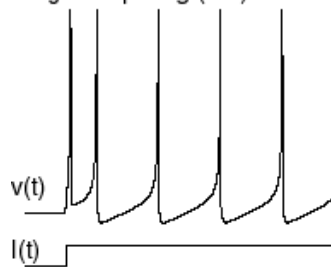
$$v' = 0.04v^2 + 5v + 140 - u + I$$

$$u' = a(bv - u)$$

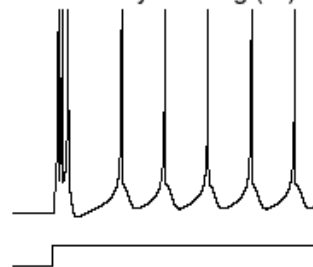
if $v = 30$ mV,
then $v \leftarrow c$, $u \leftarrow u + d$



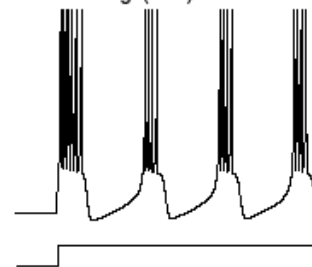
regular spiking (RS)



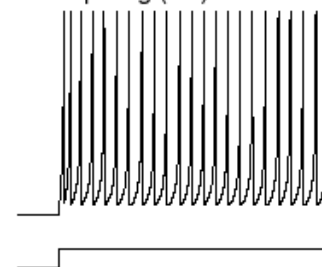
intrinsically bursting (IB)



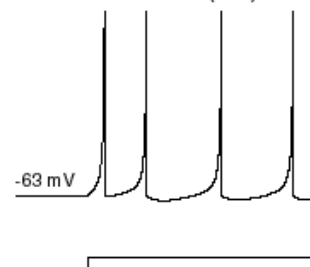
chattering (CH)



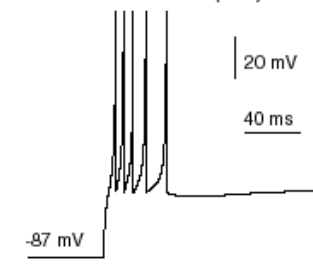
fast spiking (FS)



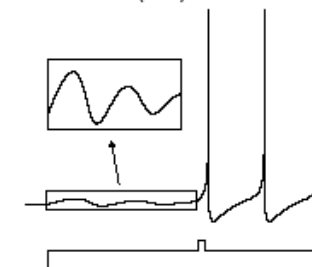
thalamo-cortical (TC)



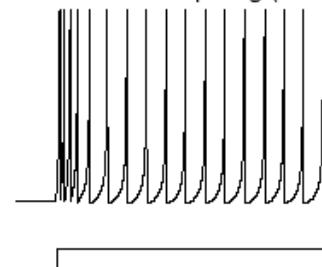
thalamo-cortical (TC)



resonator (RZ)



low-threshold spiking (LTS)



The Izhikevich Model

$$\frac{dv}{dt} = 0.04v^2 + 5v + 140 - u + \sum I_x,$$

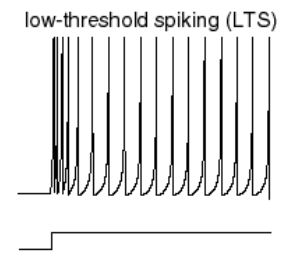
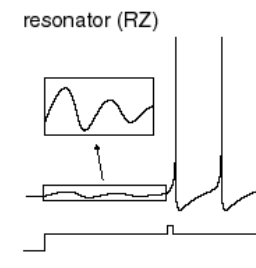
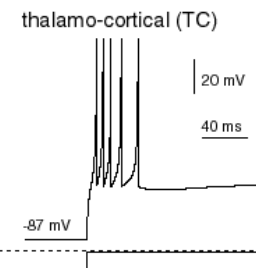
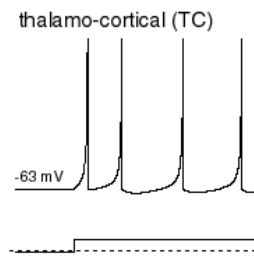
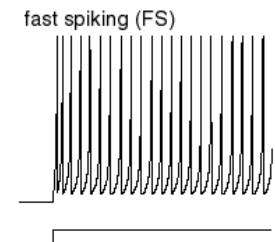
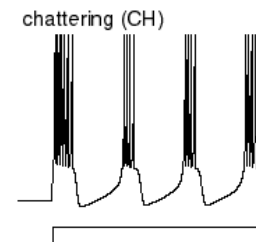
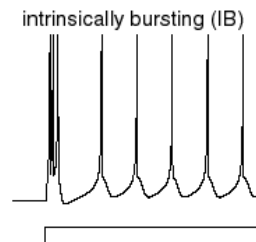
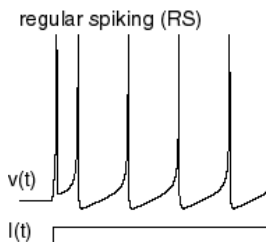
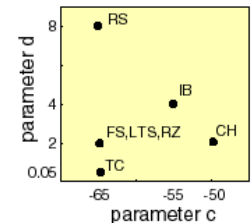
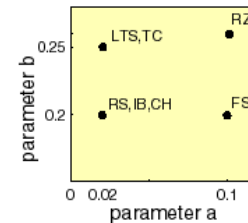
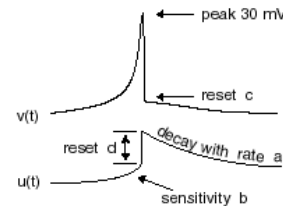
$$\frac{du}{dt} = a(bV - u).$$

- Neuronal variability

- Different spiking patterns

- Good to reproduce electrophysiological results across different species, brain regions.

$v' = 0.04v^2 + 5v + 140 - u + I$
 $u' = a(bv - u)$
 if $v = 30$ mV,
 then $v \leftarrow c$, $u \leftarrow u + d$



The Leaky integrate-and-fire (LIF) model (Lapicque 1907)

Subthreshold dynamics ($V < V_{th}$):

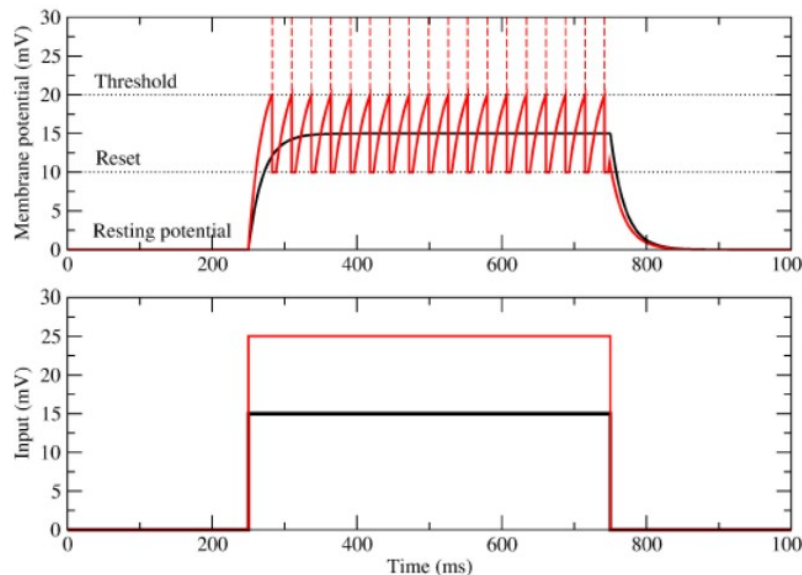
$$\tau \frac{dV}{dt} = -(V - V_{rest}) + R \cdot I$$

Spike emitted (by hand) at $t = t_{sp}$
when $V = V_{th}$

Then voltage reset to $V = V_{reset}$
(Optional) refractory period:

$$V(t) = V_{reset} \text{ for } t_{sp} < t < t_{sp} + \tau_{ref}$$

$$I_{th} = \frac{V_{th} - V_{rest}}{R}$$



Rescaling $\tau \frac{dV}{dt} = -(V - V_{rest}) + R \cdot I$:

$$v \equiv \frac{V - V_{rest}}{V_{th} - V_{rest}}; \quad i \equiv \frac{RI}{V_{th} - V_{rest}}; \quad t' \equiv t / \tau \Rightarrow$$

$$\frac{dv}{dt'} = -v(t') + i \quad \text{with threshold at } v = 1$$

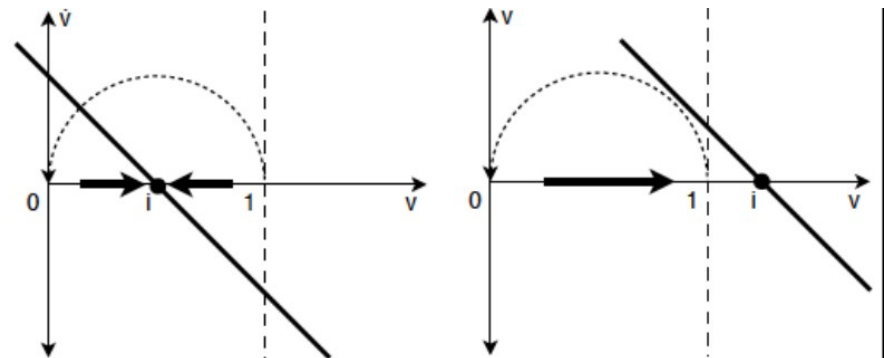
Stable fixed point at $v = i$

For $i < 1$ the membrane potential goes to the fixed point and stays there (no spikes)

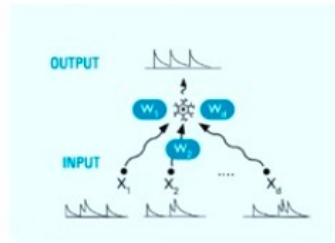
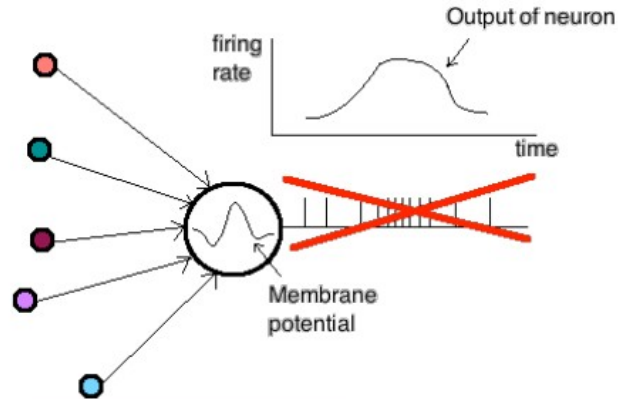
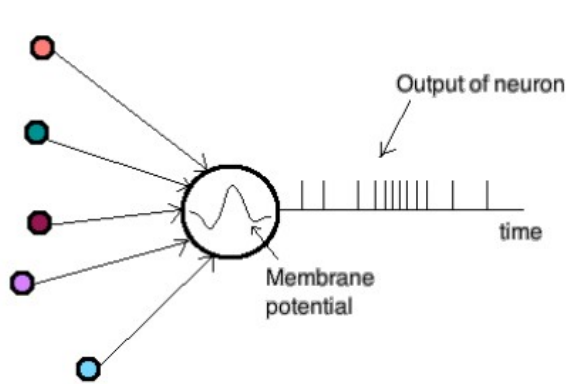
For $i > 1$ the membrane potential gets to the threshold and a spike occurs

After the spike the membrane potential is reset to 0 and the process starts again

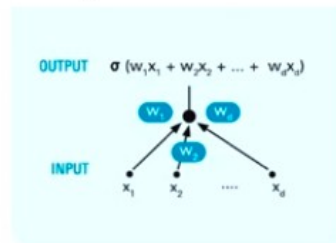
The neuron keeps firing regularly while the above threshold stimulus is on



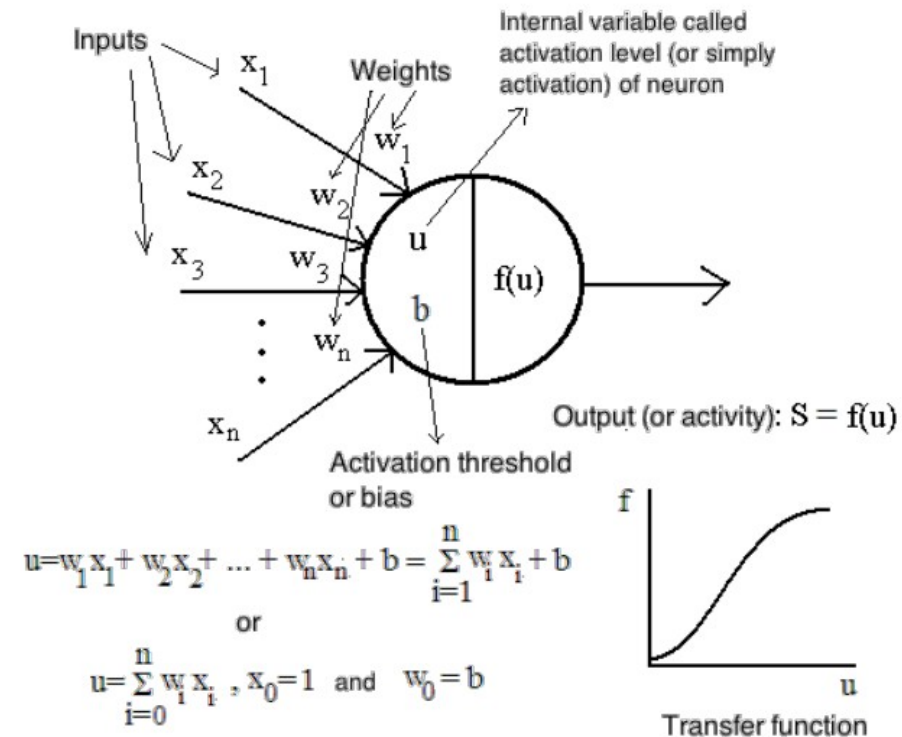
Spiking versus firing rate models



Spiking model



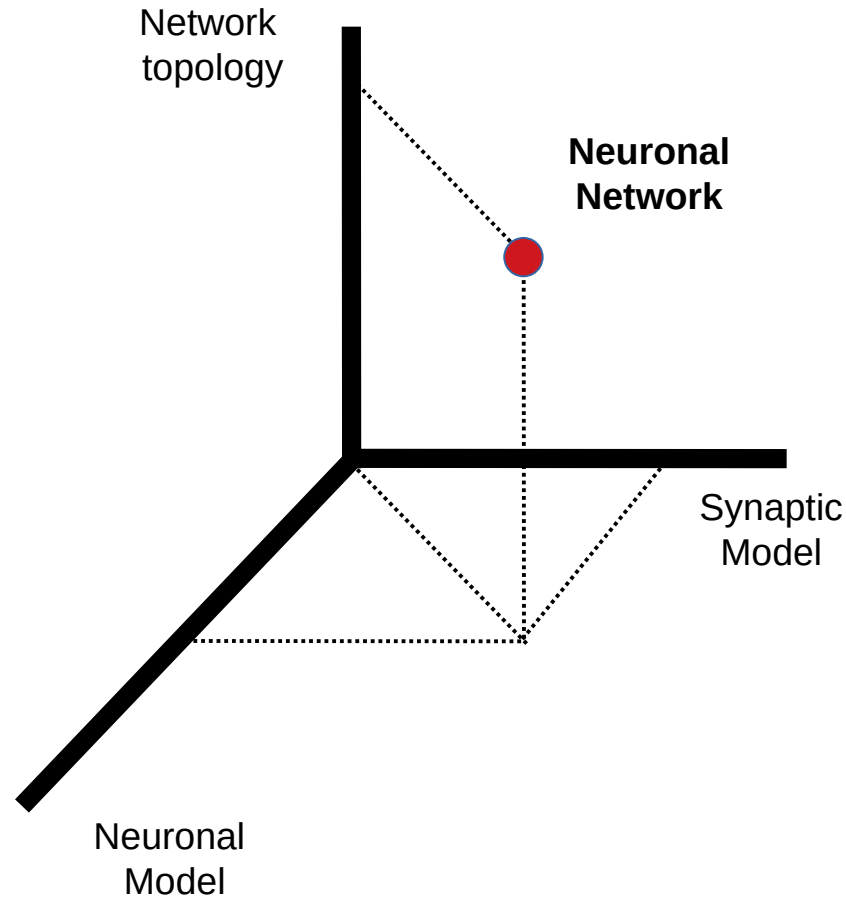
Firing rate model



Firing rate models are among the earliest forms of neuron modeling (late 1930s)

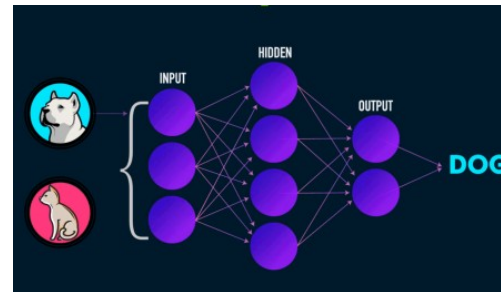
- They are the default neuron model used by Artificial Neural Networks (ANNs)
- In brain modeling, firing rate models are supposed to mimic not single cells but the "average" firing behavior of cell populations

Biological Neuronal Network

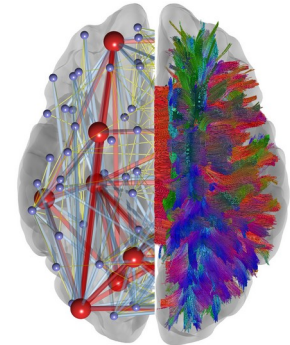


First part: chose the neuronal model (check!)

Can we build a machine that is similar to the brain?



Artificial Neural
Networks (ANN)



Biological Neural
Networks (BNN) Models

Can we build a machine that can do a specific task?

How the brain does a specific task?



Read the following text

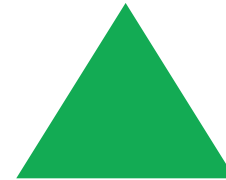
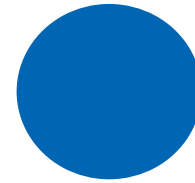
I find it rea*l* amazin* t*at y*u ar*
abl* to re*d t*is tex* despit* th* fac*
*hat more t*an t*ent* perc*n* of t**
char*cte*s a*e mis*ing.
his mean t*at you* brai* i* abl* ** fill
in missin* info*matio*.



Can we build a machine that can do a specific task? How the brain does a specific task?



Tell me the color of each symbol



**Can we build a machine that can do a specific task?
How the brain does a specific task?**



Tell me the color of each word

Red

Blue

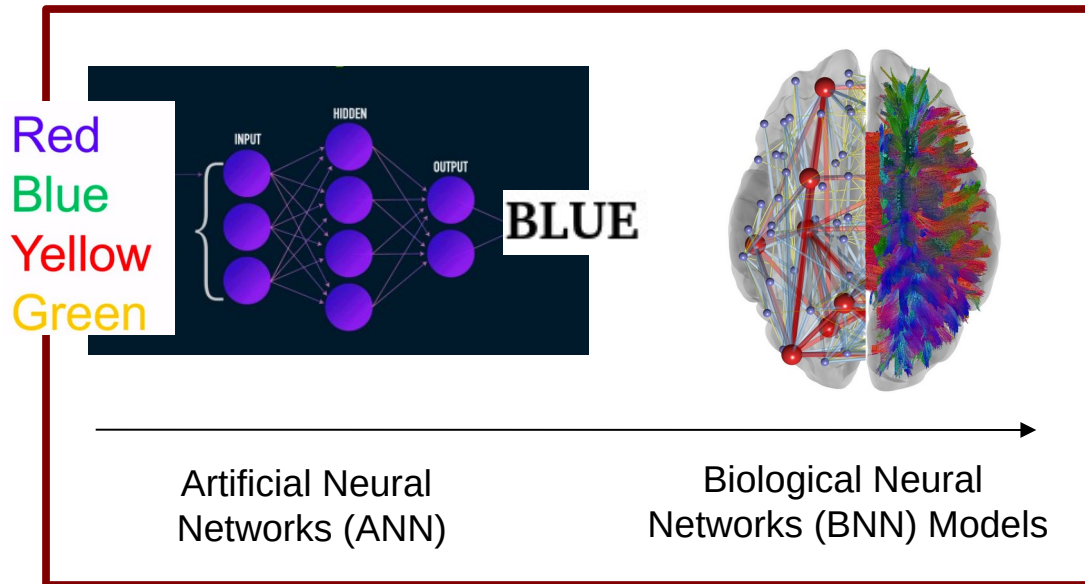
Yellow

Green



Can we build a machine that can do a specific task? How the brain does a specific task?

Tell me the color of each word



Simplified Models
1D or 2D Models
Firing rate

HH Model

More biological ingredients
Modified HH Models
Compartmental Models

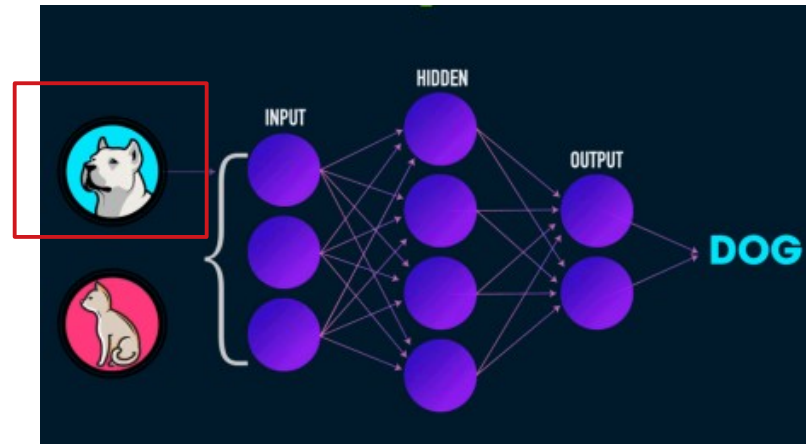
- Leaky integrate-and-fire (LIF) model;
- Non-linear LIF (quadratic, exponential);
- Adaptive exponential integrate-and-fire (AdEx) model
- Izhikevich model; Fitz-Hugh Nagumo model

Can we build a machine that can do a specific task?

How the brain does a specific task?



Tell me if it is a dog or a cat

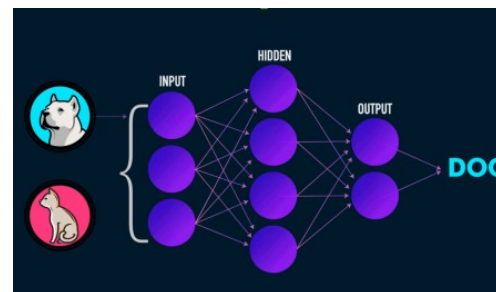
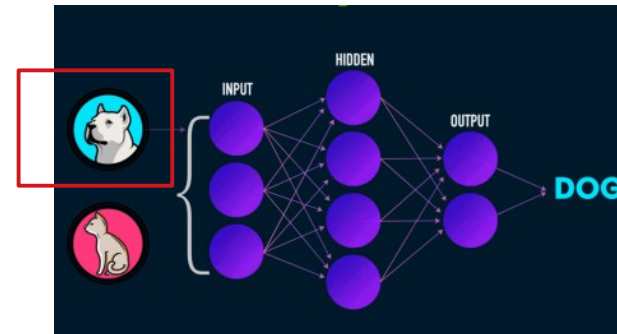


Can we build a machine that can do a specific task?

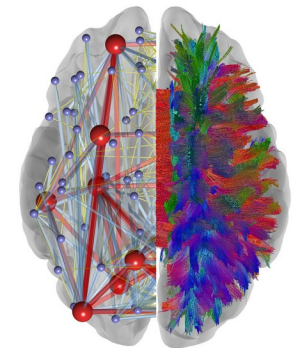
How the brain does a specific task?



Tell me if it is a dog or a cat



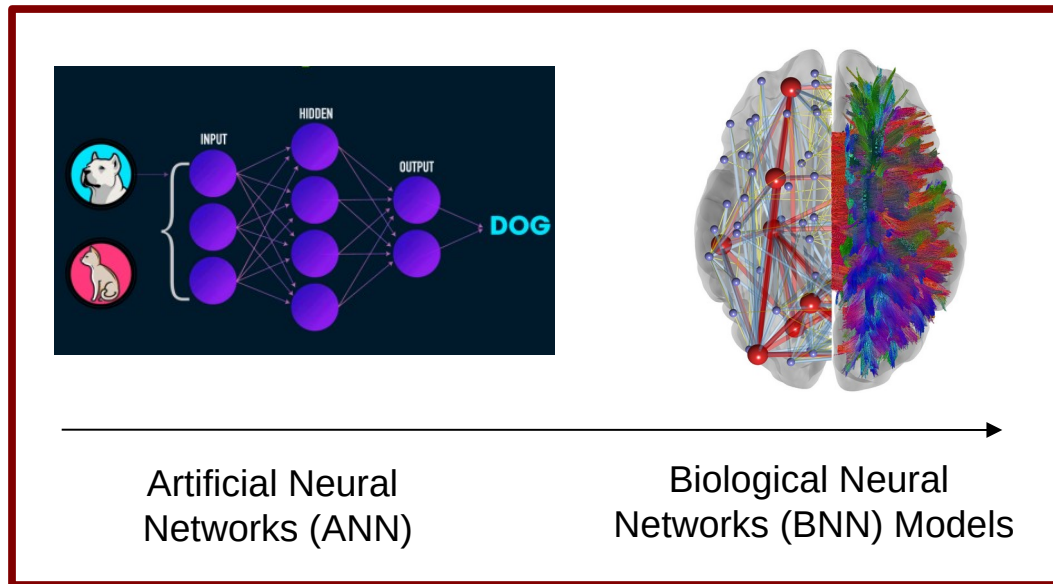
Artificial Neural
Networks (ANN)



Biological Neural
Networks (BNN) Models

Can we build a machine that can do a specific task?

How the brain does a specific task?



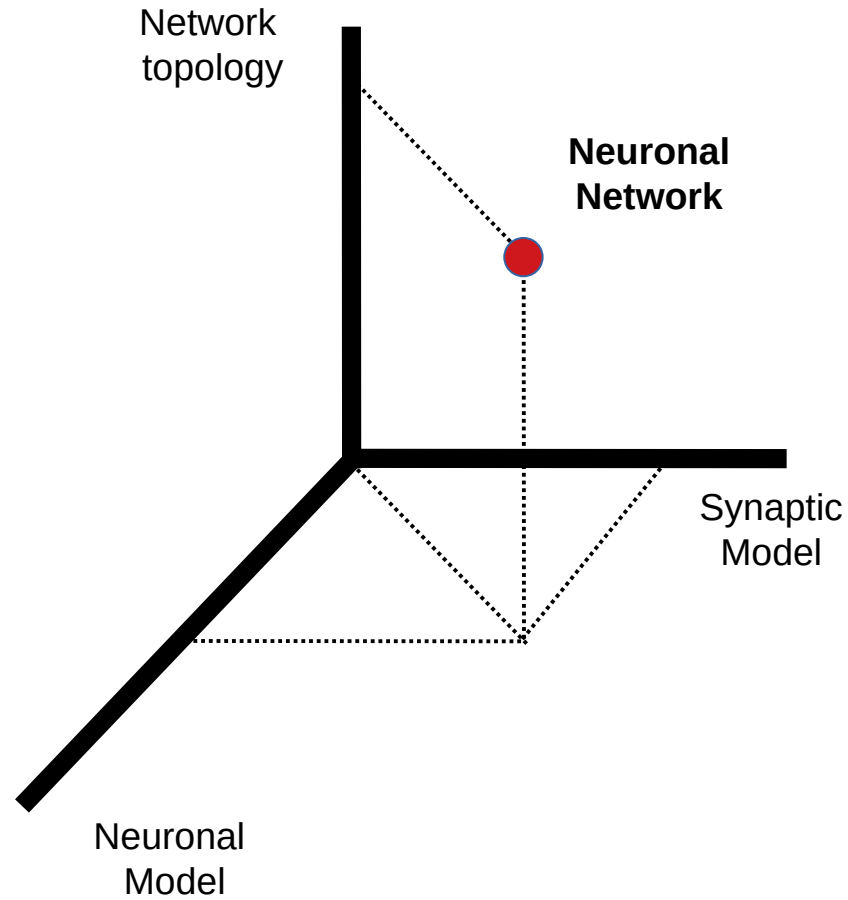
Simplified Models
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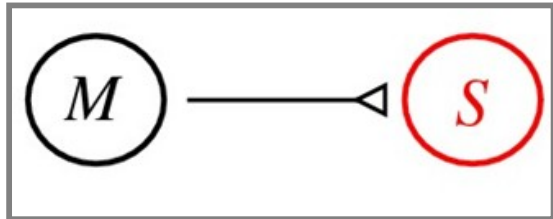
Biological Neuronal Network



Synaptic Models

How can we connect two neurons?

Hodgkin-Huxley neuron model: $C_m \frac{dV}{dt} = \overline{G}_{Na} m^3 h (E_{Na} - V) + \overline{G}_K n^4 (E_K - V) + G_m (V_{rest} - V) + I + \sum I_{syn}$

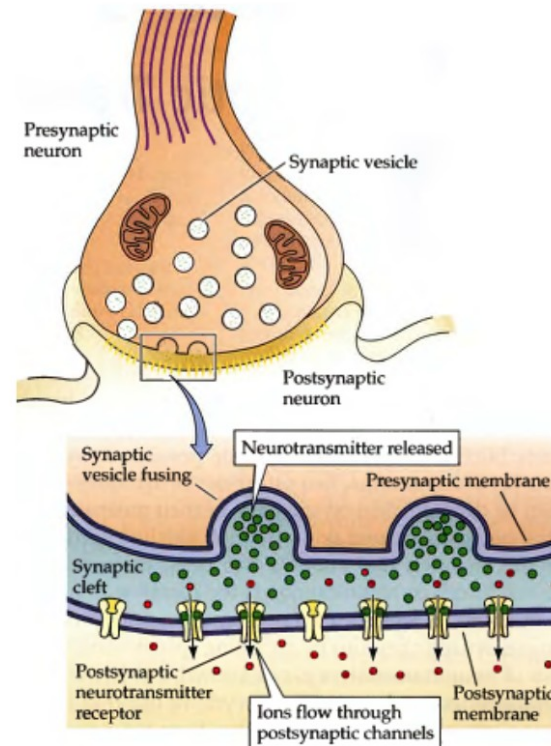
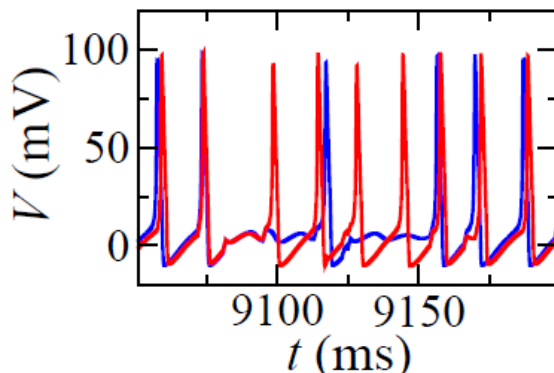


$$\frac{dx}{dt} = \alpha_x(V)(1 - x) - \beta_x(V)x$$

AMPA and GABA_A Synapses:

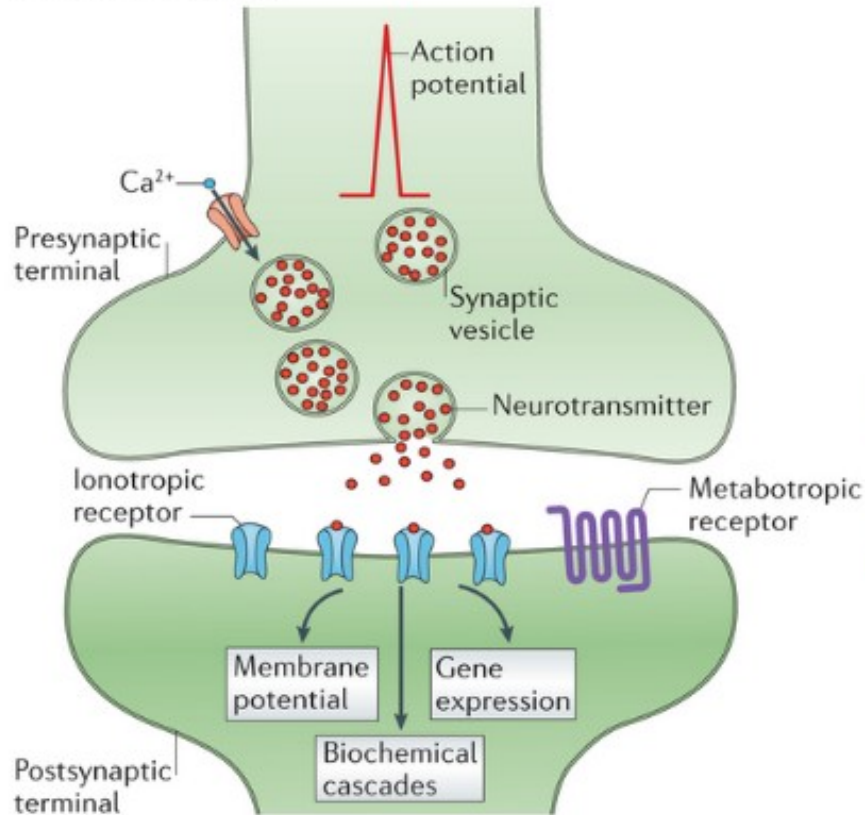
$$I^{(i)} = g_i r^{(i)} (V - E_i)$$

$$\frac{dr^{(i)}}{dt} = \alpha_i [T] (1 - r^{(i)}) - \beta_i r^{(i)}$$

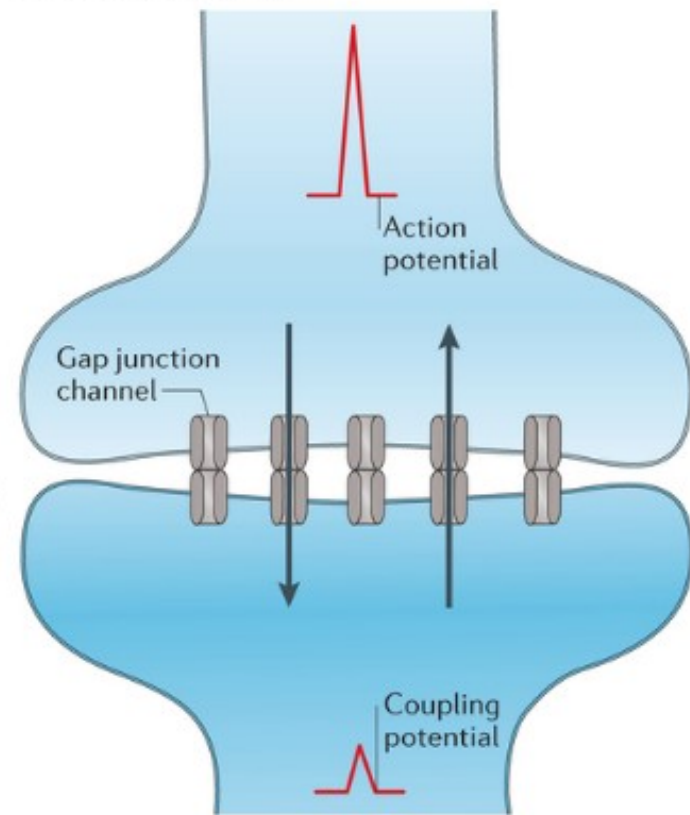


Chemical versus Electrical synapses

a Chemical synapse

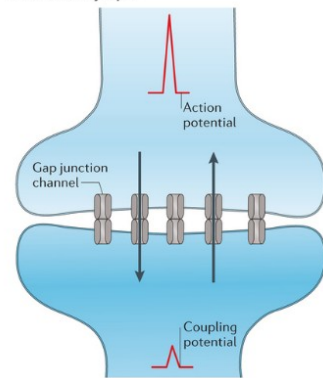


b Electrical synapse

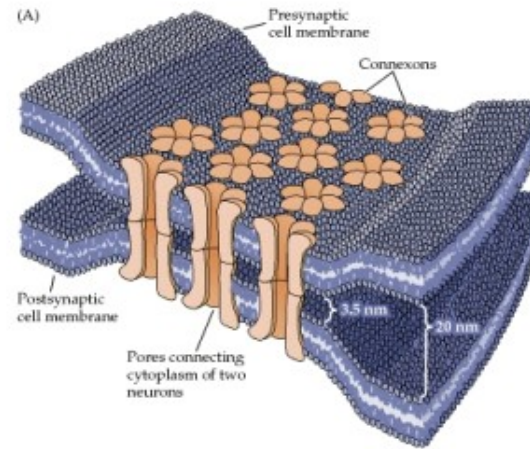


Electrical synapse

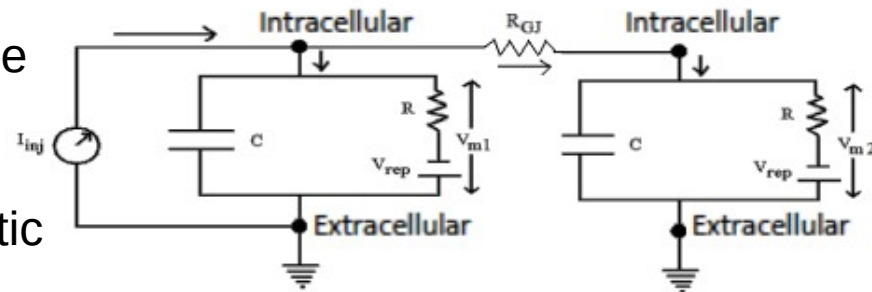
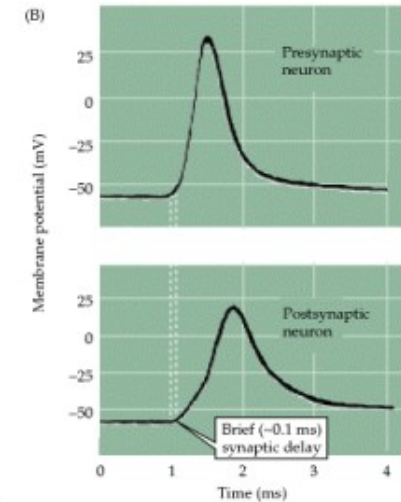
b Electrical synapse



- 2 neurons linked together by gap junctions
- Rapid and bidirectional communication
- Excitation and inhibition at the same synapse
- Occur between neurons and glia
- An action potential in the presynaptic neuron causes the postsynaptic neuron to be depolarized within a fraction of Millisecond
- Gap junction coupling can be modeled by a single resistance connecting the 2 cells



Neuroscience. 2nd edition.
Purves D, Augustine GJ, Fitzpatrick D, et al., editors.
Sunderland (MA): Sinauer Associates, 2001

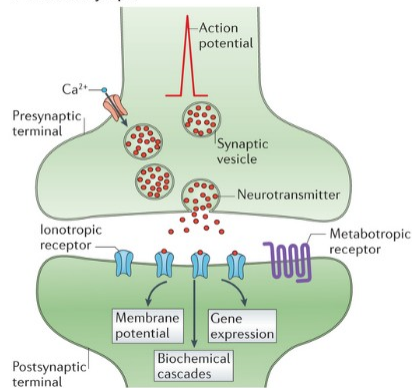


$$V_2 = k V_1$$

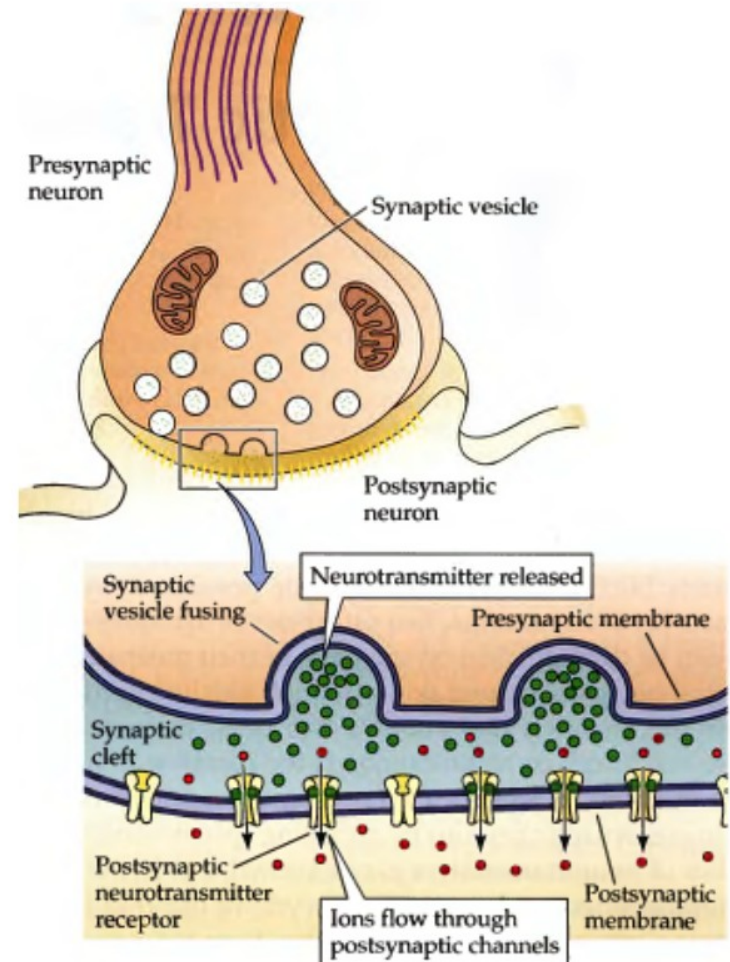
$$I_{GJ} = G_{GJ} (V_1 - V_2)$$

Chemical synapse

a Chemical synapse

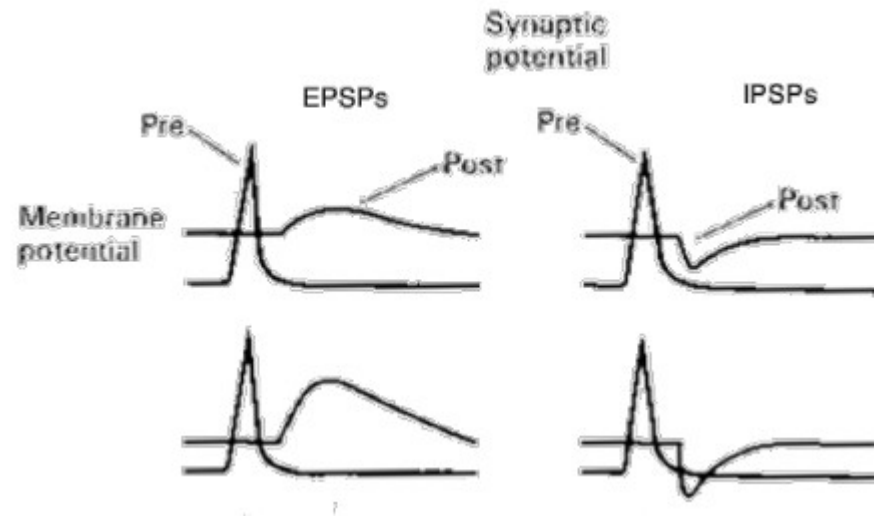
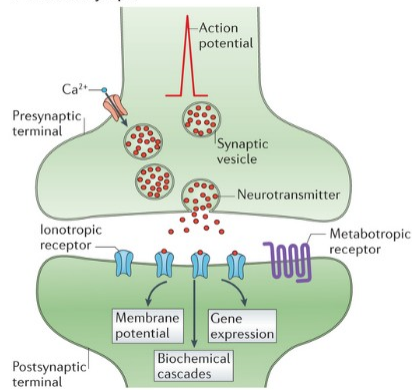


- Signal transduction
- Excitatory or inhibitory
- Slower and unidirectional communication
- Ionotropic (fast) and metabotropic (slow)
- Many neurotransmitters, but the most common in the cortex are:
 - Glutamate (usually excitatory)
 - γ -aminobutyric acid (GABA) (usually inhibitory)
- Dynamics depends on receptor type:
 - Glutamate receptors: AMPA/Kainate and NMDA
 - GABA receptors: GABA A and GABA B
- Short-term and long-term synaptic plasticity



Post synaptic potential

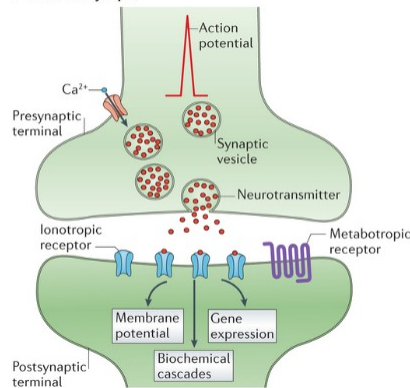
a Chemical synapse



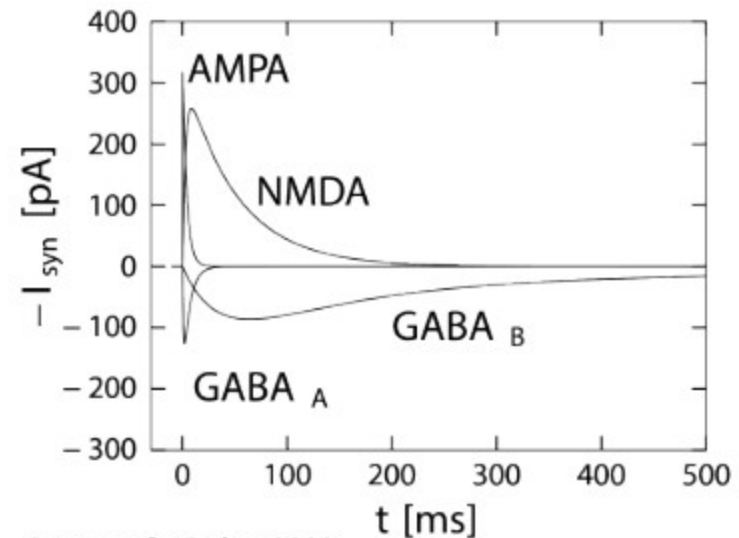
- Excitatory postsynaptic potential (**EPSP**): Transient **depolarization** of the postsynaptic membrane by presynaptic release of neurotransmitter
- Inhibitory postsynaptic potential (**IPSP**): Transient **hyperpolarization** of the postsynaptic membrane by presynaptic release of neurotransmitter

Synaptic models

a Chemical synapse



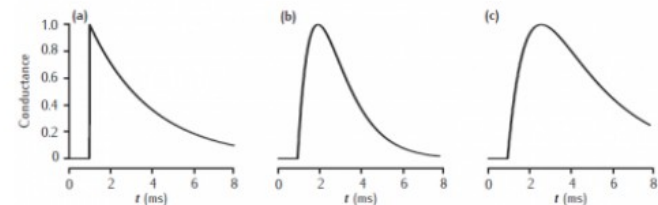
$$I^{(i)} = g_i r^{(i)} (V - E_i)$$



Gerstner & Kistler, 2002

- There are many models, the most common in network models assumes a synaptic current $I_{\text{syn}} = g(t) (V_{\text{post}} - V_{\text{rev}})$
- $V_{\text{rev}} = -75 \text{ mV}$ (inhibitory synapses) and $V_{\text{rev}} = 0$ (excitatory synapses)
- $g(t)$ = synap/c conductance of postsynaptic cell
- Synaptic delays can also be introduced

- The time course of $g(t)$ can be modeled by kinetic equations or fixed time course functions are used



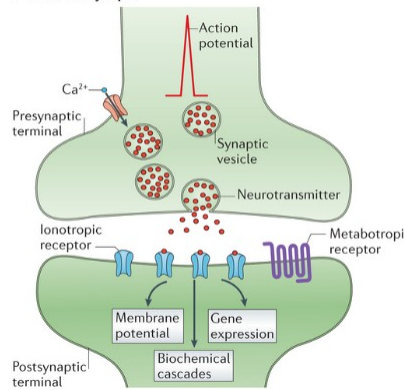
$$(a) \quad g(t) = \bar{g} \exp\left(-\frac{t - t_j}{\tau}\right)$$

$$(b) \quad g(t) = \bar{g} \frac{t - t_j}{\tau} \exp\left(-\frac{t - t_j}{\tau}\right) \quad (\text{alpha function})$$

$$(c) \quad g(t) = \bar{g} \frac{\tau_1 \tau_2}{\tau_1 - \tau_2} \left(\exp\left(-\frac{t - t_j}{\tau_1}\right) - \exp\left(-\frac{t - t_j}{\tau_2}\right) \right)$$

Synaptic models

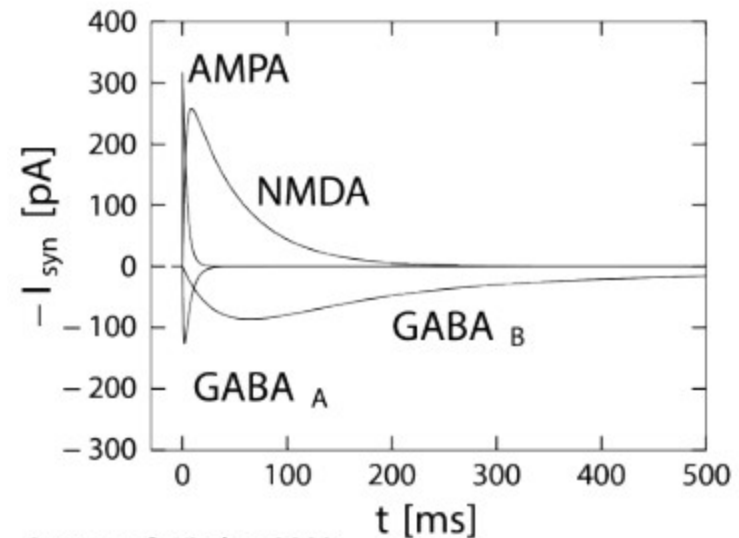
a Chemical synapse



$$I^{(i)} = g_i r^{(i)} (V - E_i)$$

$$\frac{dr^{(i)}}{dt} = \alpha_i [T] (1 - r^{(i)}) - \beta_i r^{(i)}$$

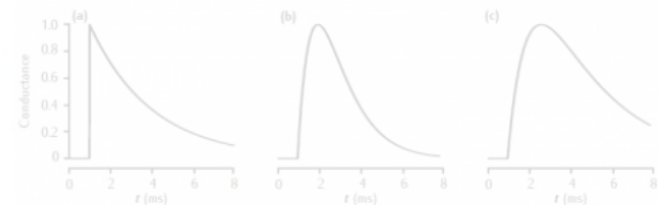
$$n \xrightleftharpoons[\alpha_n]{\beta_n} 1 - n$$



Gerstner & Kistler, 2002

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Synaptic Plasticity

Biological Neuronal Network

Network
topology

Neuronal
Network

Synaptic
Model

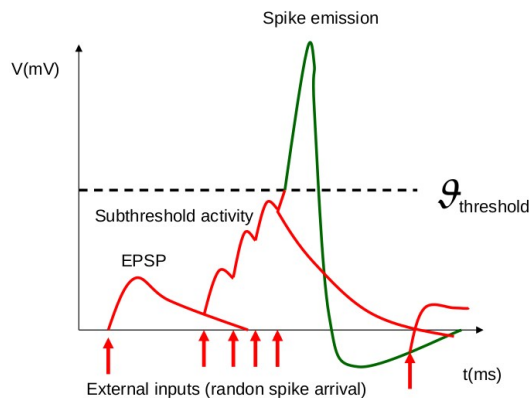
$$I^{(i)} = g_i r^{(i)} (V - E_i)$$

$$\frac{dr^{(i)}}{dt} = \alpha_i [T] (1 - r^{(i)}) - \beta_i r^{(i)}$$

Neuronal
Model

$$C_m \frac{dV}{dt} = \overline{G}_{Na} m^3 h (E_{Na} - V) + \overline{G}_K n^4 (E_K - V) + G_m (V_{rest} - V) + I + \sum I_{syn}$$

$$\frac{dx}{dt} = \alpha_x(V)(1 - x) - \beta_x(V)x$$



Synaptic plasticity, learning, and memory

- Synaptic plasticity is the ability of the synapses to **change their conductances/weights (g) over time** in response to activity or experience.

- It is considered the basic phenomenon underlying learning and memory

Synaptic potentiation: strengthening of synapses $g \rightarrow g+A$
Synaptic depression: weakening of synapses $g \rightarrow g-A$

- Hebb (1949): if input from neuron A often contributes to the firing of neuron B, then the synapse from A to B should be strengthened. “Neurons that fire together wire together”

- More recently, Hebb’s suggestion has been generalized to include decreases in strength arising from repeated failure of neuron A to be involved in the activation of neuron B.

$$I^{(i)} = g_i r^{(i)} (V - E_i)$$
$$\frac{dr^{(i)}}{dt} = \alpha_i [T] (1 - r^{(i)}) - \beta_i r^{(i)}$$

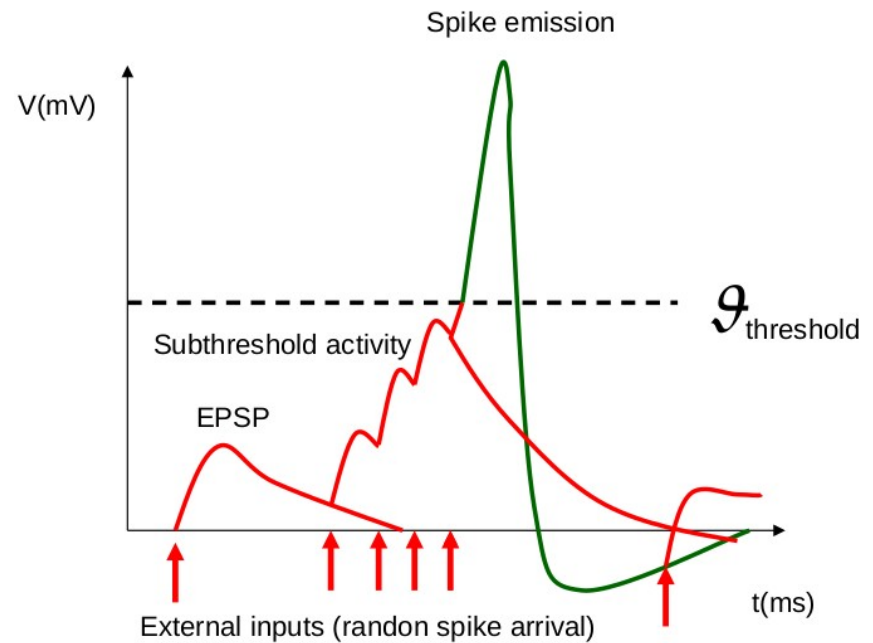
Synaptic plasticity, learning, and memory

$$I^{(i)} = g_i r^{(i)} (V - E_i)$$

$$\frac{dr^{(i)}}{dt} = \alpha_i [T] (1 - r^{(i)}) - \beta_i r^{(i)}$$

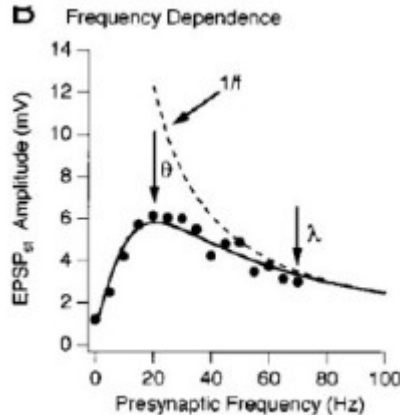
Potential: strengthening of synapses
should increase the peak of the EPSP

Depression: weakening of synapses
Should decrease the peak of the EPSP

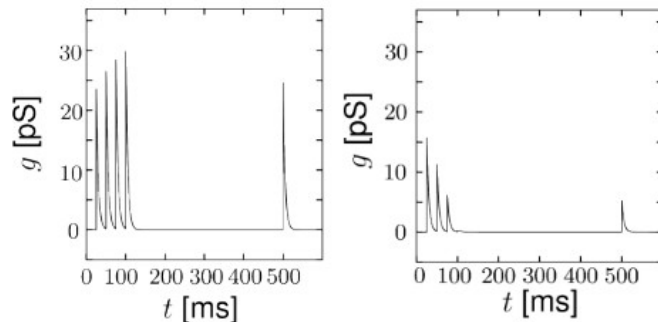


Short term Plasticity

-Facilitation of excitatory cortical synapse
in a slice of rat somatosensory cortex



-Frequency dependence
of the facilitation



Simulation of facilitation (left) and depression (right):
sequence of 4 presynaptic spikes followed by a 5th
spike 400 ms after (Gerstner & Kistler, 2002)

Synaptic conductance given by

$$g(t) = \bar{g} p_s p_{rel}$$

p_s = probability that a postsynaptic channel opens given that a transmitter was released by presynaptic neuron $\rightarrow$ modeled by an α function

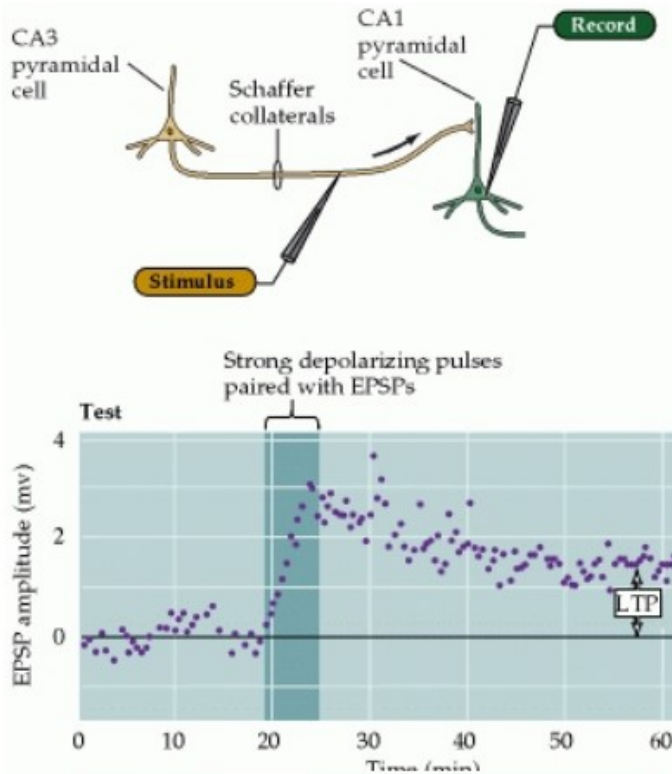
p_{rel} = probability that a transmitter is released by presynaptic neuron following the arrival of an action potential

p_{rel} is affected by synaptic **facilitation** and **depression**

$$\tau_p \frac{dp_{rel}}{dt} = p_0 - p_{rel} + f_F (1 - p_{rel}) \sum_j \delta(t - t_j)$$

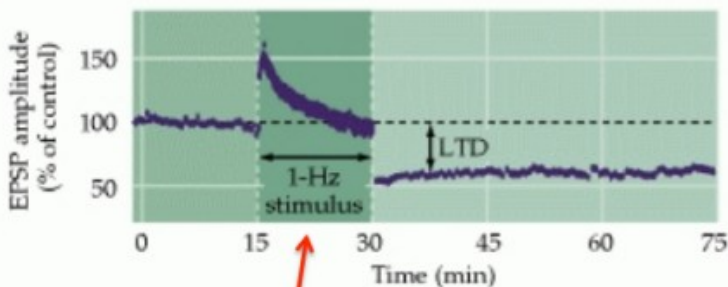
$$\tau_p \frac{dp_{rel}}{dt} = p_0 - p_{rel} - f_D p_{rel} \sum_j \delta(t - t_j)$$

Long-term potentiation and depression



- Models of long-term plasticity

$$I_{\text{syn}} = g \cdot s(t) \cdot (V_{\text{post}} - V_{\text{rev}})$$
- Changes in g (maximal synaptic conductance) to model incorporation or removal of channels
- Implemented by hand or through a learning rule (e.g. Hebb's rule)
- The time-scale of changes in g is much slower than the one of membrane potential dynamics



Low-frequency stimulation

Purves et al. (2001)

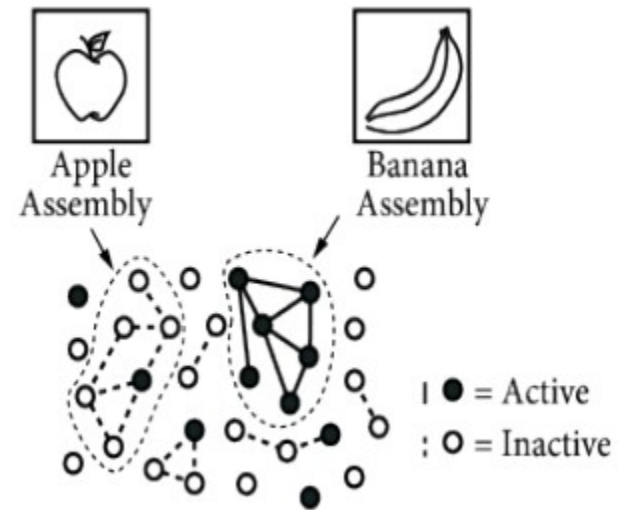
- Long term potentiation (LTP)
- Long term depression (LTD)

$g \rightarrow g + \Delta$ (LTP)

$g \rightarrow g - \Delta$ (LTD)

Hebbian rules and Learning

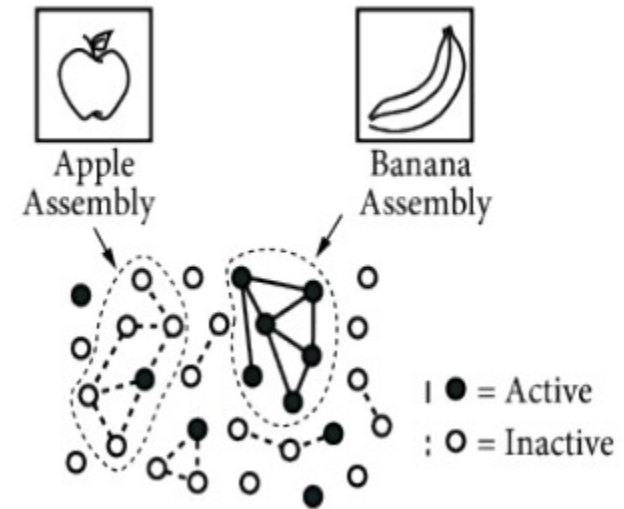
- Mechanism of synaptic weight change proposed by Hebb in 1949 to implement formation of cell assemblies and learning
- The weight of the synapse between two neurons grows whenever their firing activity is correlated: neurons that fire together wire together
- Morris (1986) showed that LTP was necessary for the formation of memories in vivo.
- Animals where the NMDA receptors were blocked were unable to learn the position of the platforms where they could stand without swimming. And in vitro analysis showed that LTP was indeed blocked.
- Tonegawa (1996) showed that genetically altered mice with impaired NMDA receptors showed poor spatial navigation capabilities.



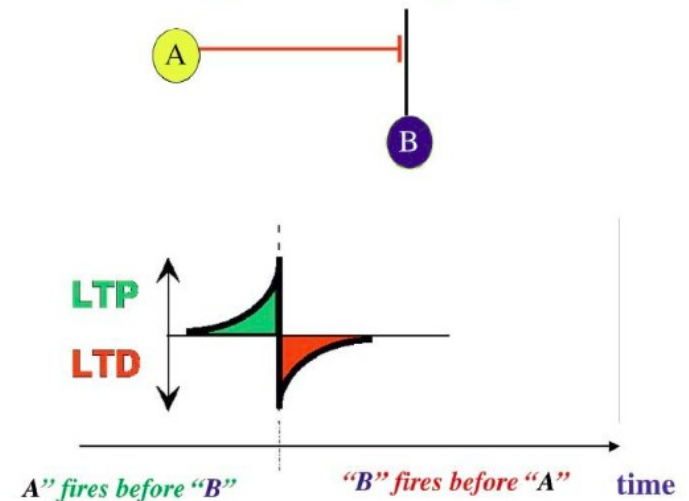
Spiking-time dependent plasticity (STDP)

- **Spike-timing dependent plasticity (STDP)**: experimental protocol of synaptic plasticity induction discovered by Markram in 1995.

The spike of pre-synaptic neuron must precede the spike of post-synaptic neuron for their weight to increase (LTP), otherwise the weight decreases (LTD)



Learning rule at the synapse



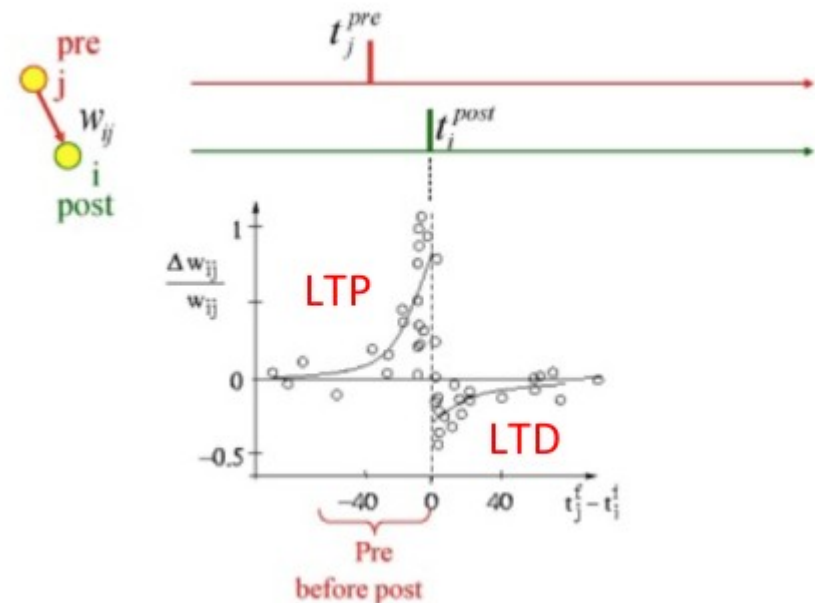
Spiking-time dependent plasticity (STDP)

- **Spike-timing dependent plasticity (STDP)**: experimental protocol of synaptic plasticity induction discovered by Markram in 1995.

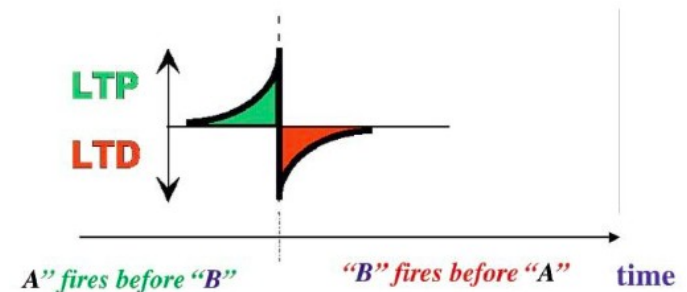
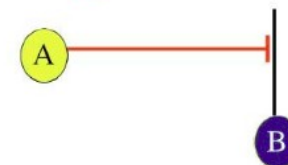
The spike of pre-synaptic neuron must precede the spike of post-synaptic neuron for their weight to increase (LTP), otherwise the weight decreases (LTD)

$$I^{(i)} = g_i r^{(i)} (V - E_i)$$

$$g = \begin{cases} g + A_+ \exp(-t/\tau_+), & \text{if } t > 0 \quad (\text{additive LTP}) \\ g - A_- g \exp(t/\tau_-), & \text{if } t < 0 \quad (\text{multiplicative LTD}) \end{cases}$$



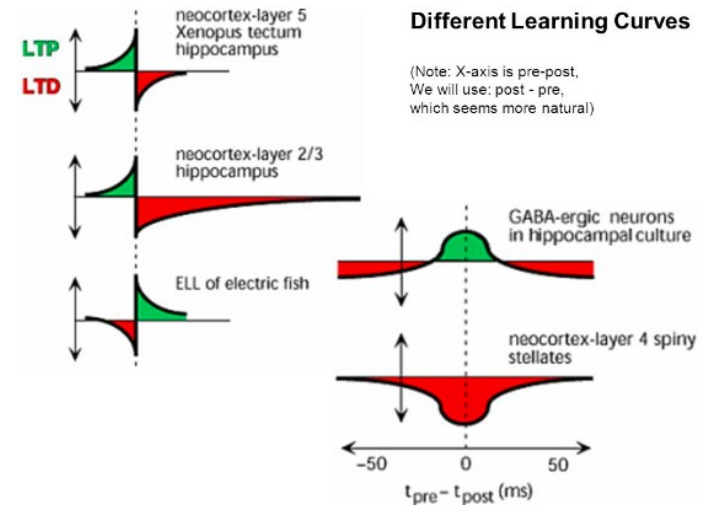
Learning rule at the synapse



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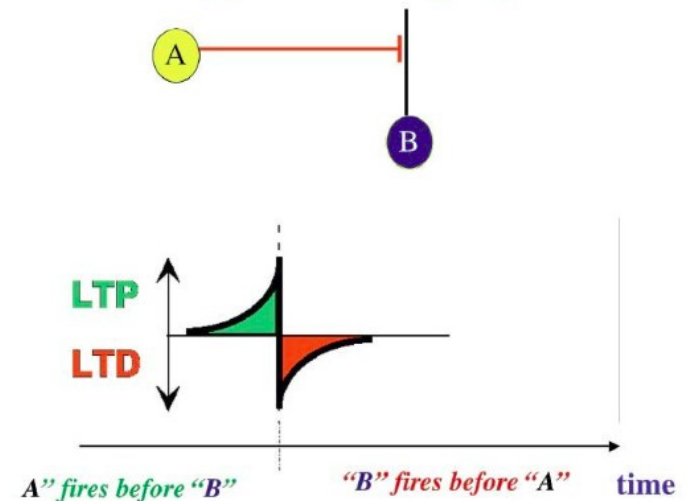
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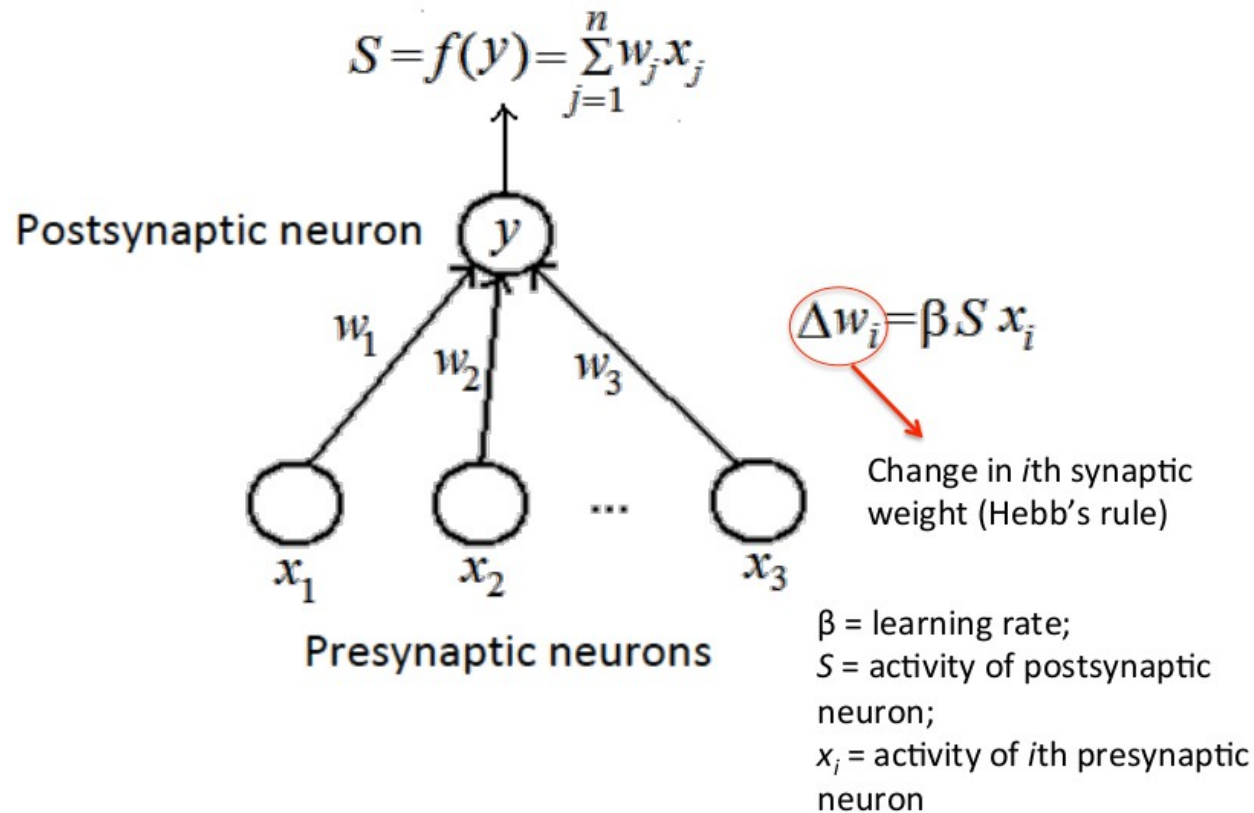
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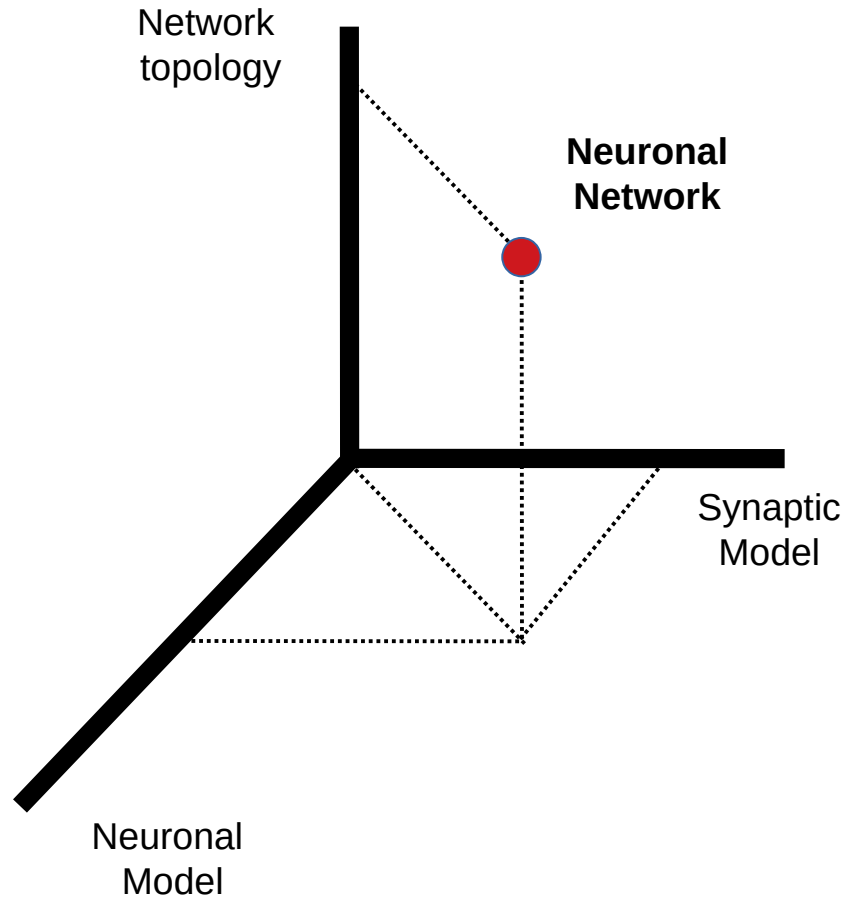
Learning rule at the synapse



Hebbian rule for firing rate based models



Biological Neuronal Network

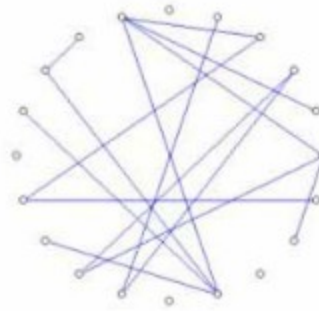


Network topology

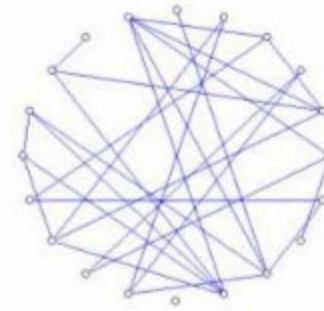
Types of architecture: Artificial x Biologically inspired (data-driven)



$p = 0$
(a)

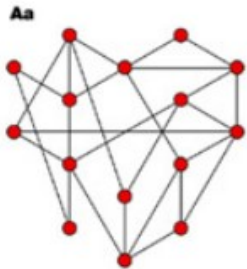


$p = 0.1$
(b)

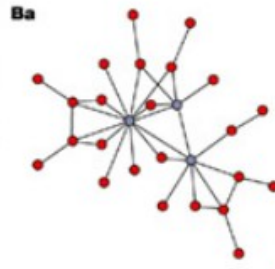


$p = 0.2$
(c)

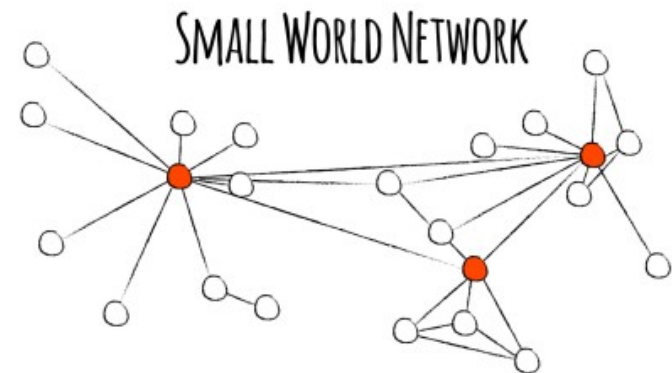
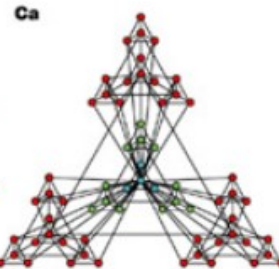
A Random network



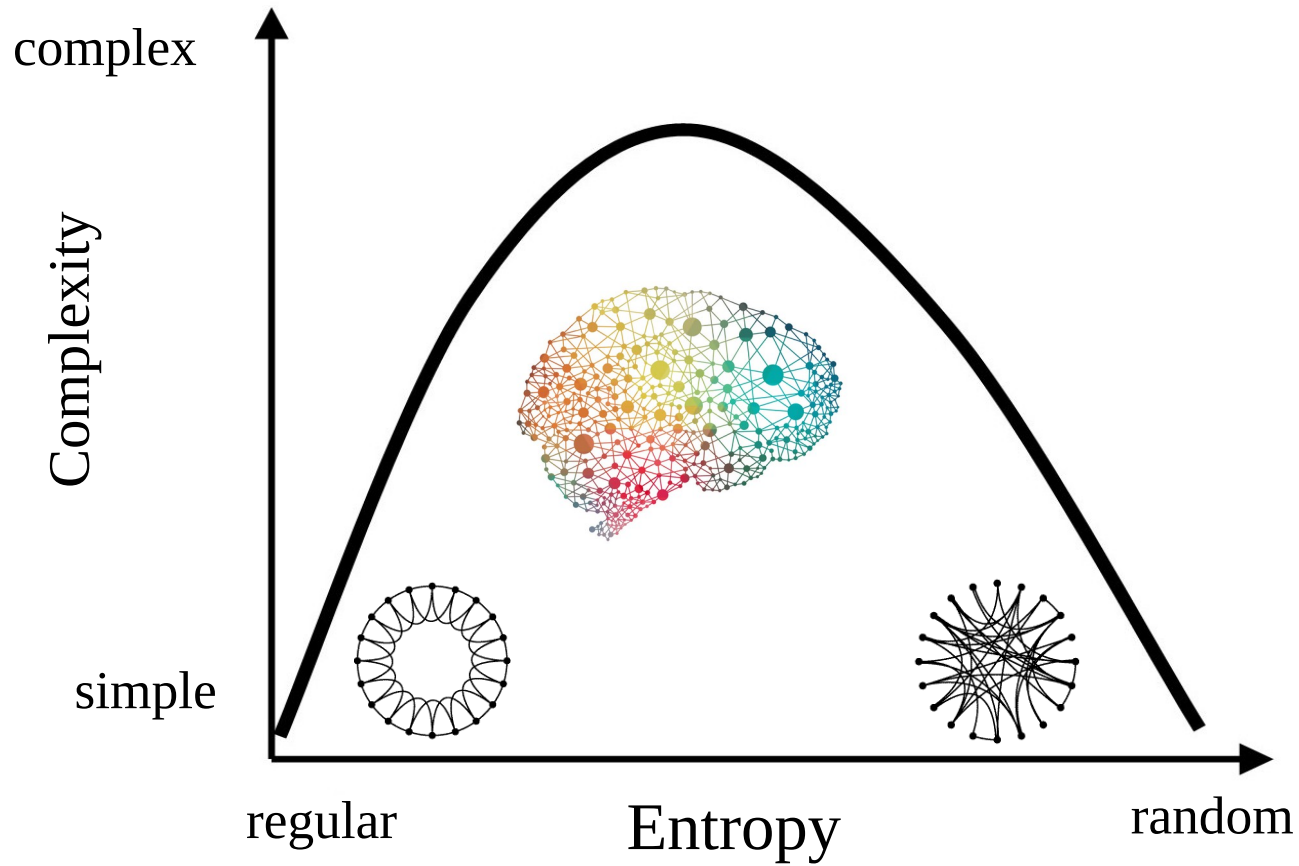
B Scale-free network



C Hierarchical network

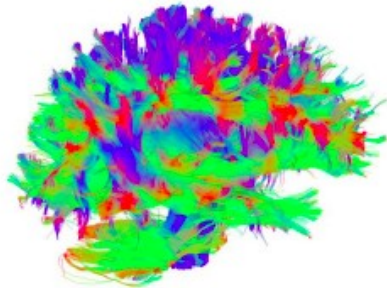


Topology

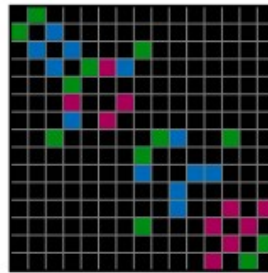


Structural Connectivity

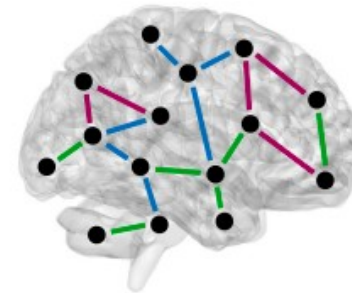
a Measurement



Example: White matter tracts (via DTI)

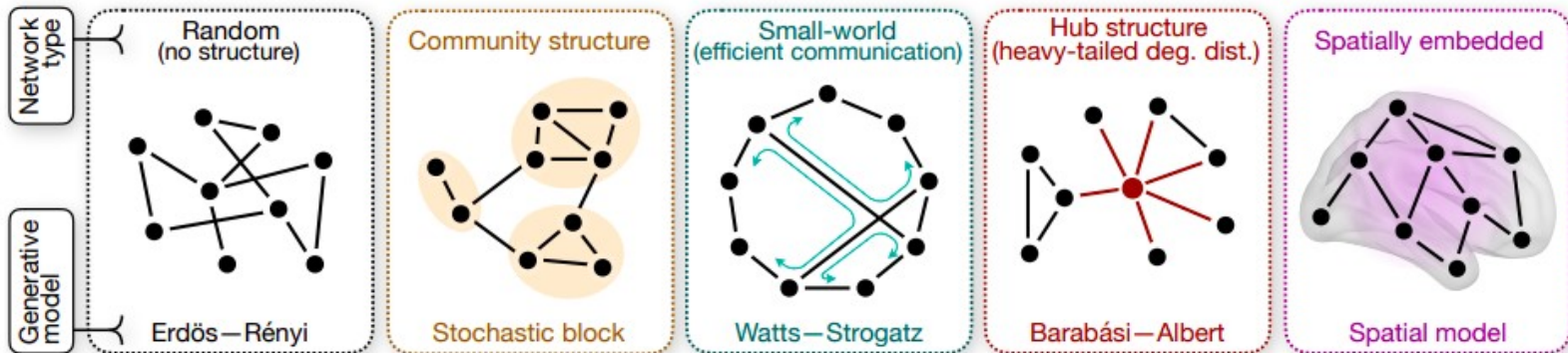


Adjacency matrix

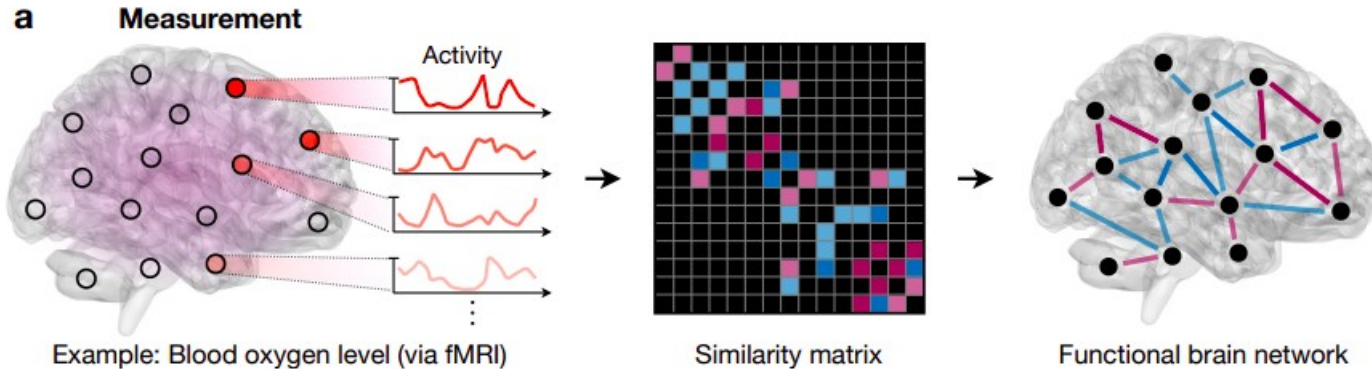


Structural brain network

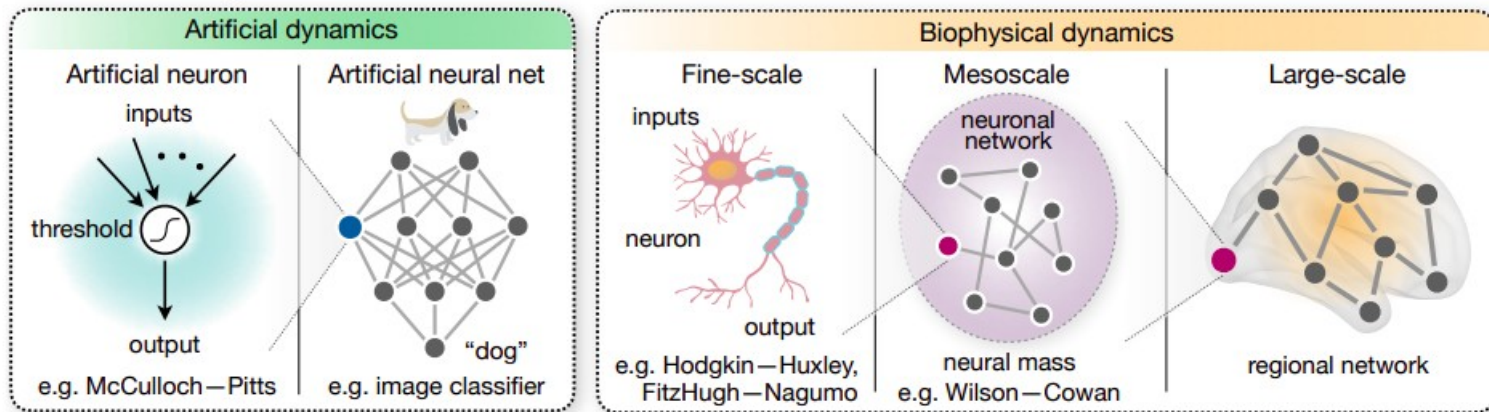
b Modeling



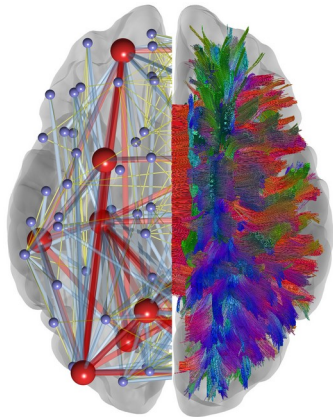
Functional Connectivity

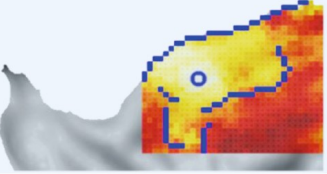
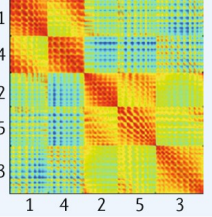
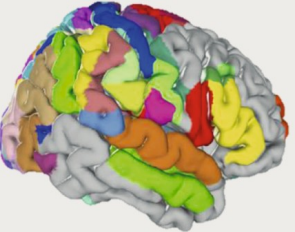
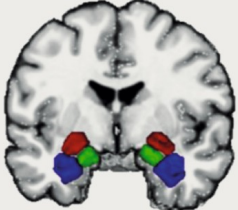
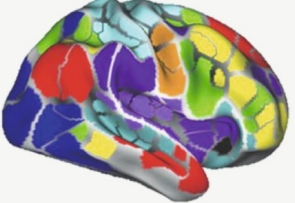
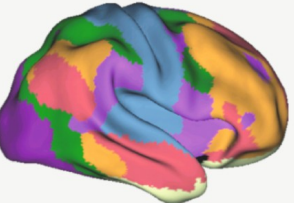


b Modeling



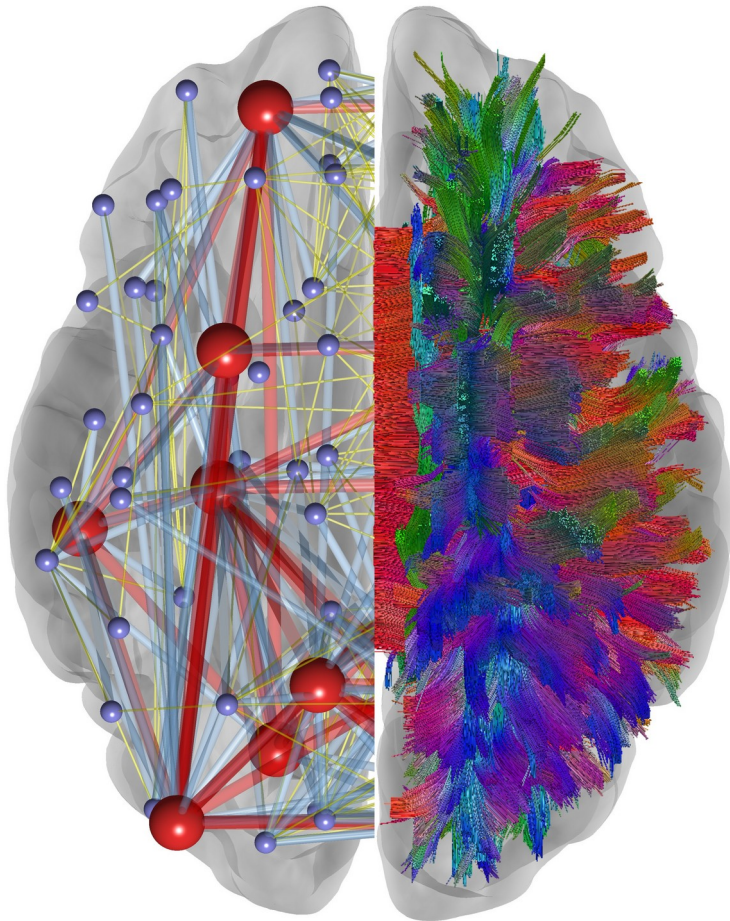
Parcelation X Integration



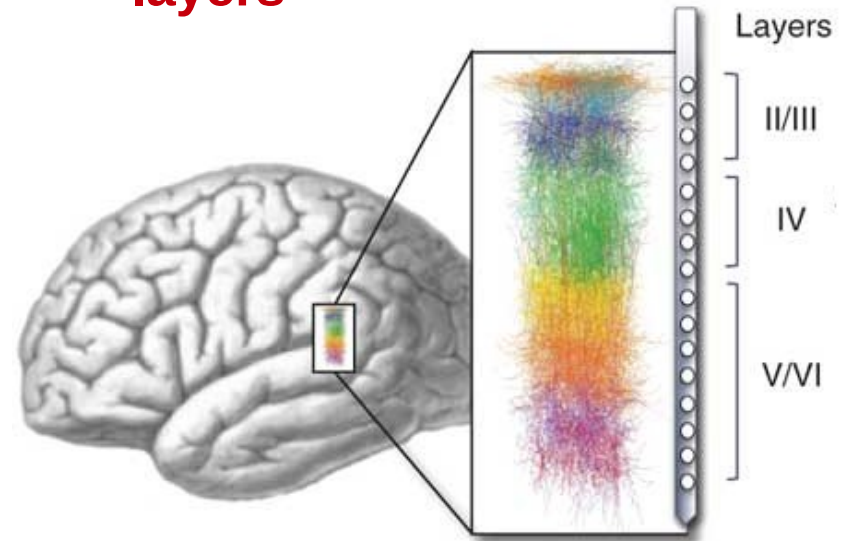
Algorithm Markers	Boundary mapping 	Clustering or factorization 
Local Histology-based: • Cytoarchitecture • Receptors • Myelin MRI-based: • Myelin • Meta-analytic activation modelling	Border detection in cortex based on cytoarchitecture 	Clustering of amygdala voxels based on their activation in behavioural paradigms 
Global MRI-based: • Resting-state functional connectivity • Meta-analytic connectivity modelling • Diffusion tractography • Structural covariance	Boundary mapping of resting-state functional connectivity of cerebral cortex 	Clustering of cerebral cortex based on resting-state functional connectivity 

Topology: Multiple scales

Macroscopic - large scale



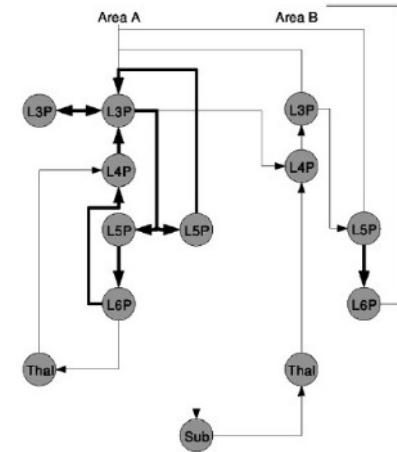
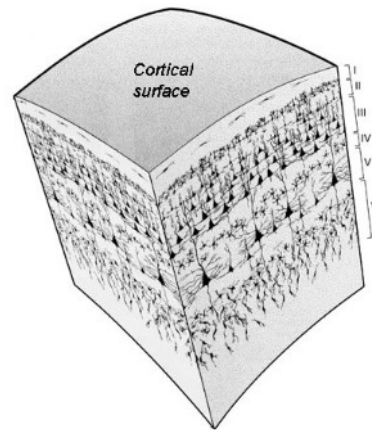
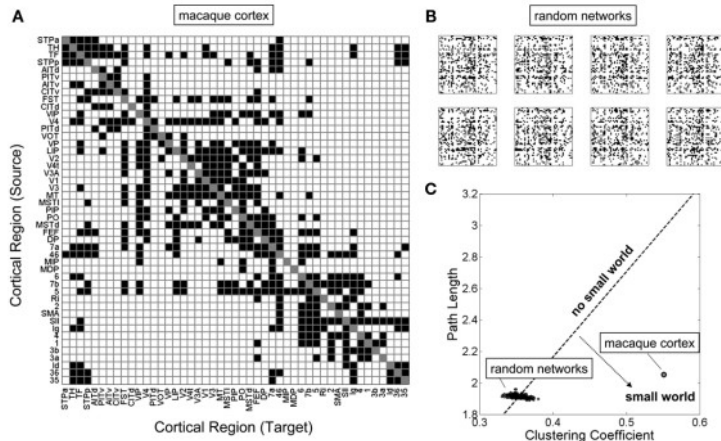
**Microscopic – local scale –
among neurons from the same
layers**



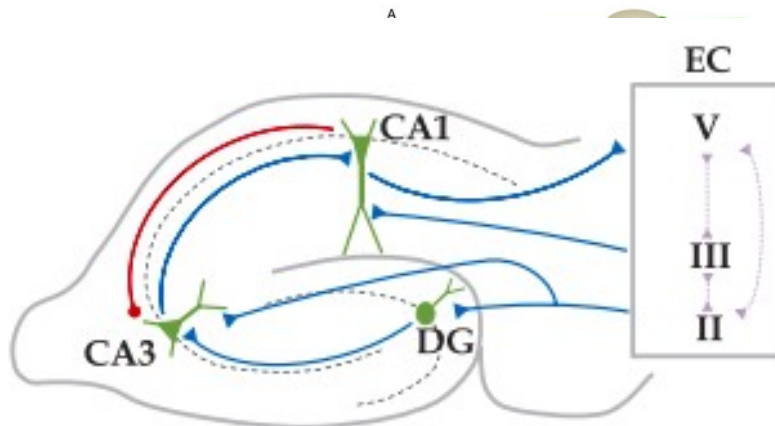
Topology

Global: connectivity among cortical areas

Local: connectivity among neurons in cortical layers/ columns



Connectivity among neurons in non-cortical structures (e.g. hippocampus dentate gyrus)



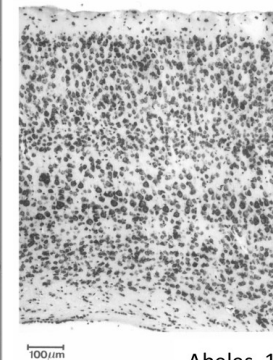
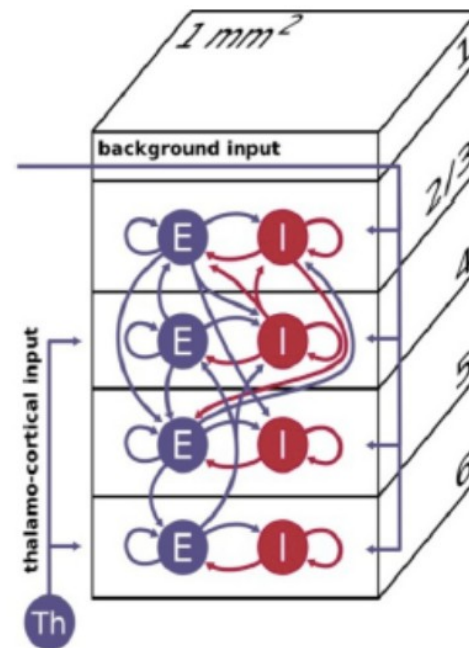
Topology microscale

- The neocortex can be subdivided into 6 layers
 - Layers differ in terms of densities and types of cells
- Local cortical microcircuit model
 - 1 mm² cortical surface
 - Integrates knowledge from over 50 experimental papers
 - Excitatory (E) and inhibitory (I) populations of point neurons in each layer
 - Layer and type-specific connection probabilities
 - Replicates well the layer-specific distribution of spike rates
- 10⁵ neurons
(80% excit. 20% inhib.)
10⁹ synapses

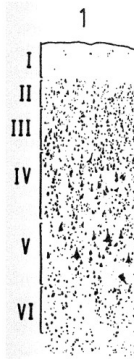
Cerebral CORTEX

The Cell-Type Specific Cortical Microcircuit: Relating Structure and Activity in a Full-Scale Spiking Network Model 3

Potjans and Diesman



Abeles, 1991



C. Elegans connectome

- Small Nervous System ~ **302** neurons
 - Genome mapped (1998)
 - Somatic connectome mapped: White et al. **1986**
 - **279** neurons;
 - ~**8000** connections: (Gap, Glutamatergic, Cholinergic, GABA)
 - Constantly being **updated** (e.g. Varshney et al 2011, Haspel et al 2012, Cook et al 2019)
 - Additional **extra synaptic currents** mapped (monoamines, neuropeptides)
 - **Evolutionary** Information is available
 - Central resource combine the data: **Wormatlas**
- + more

Thank you!



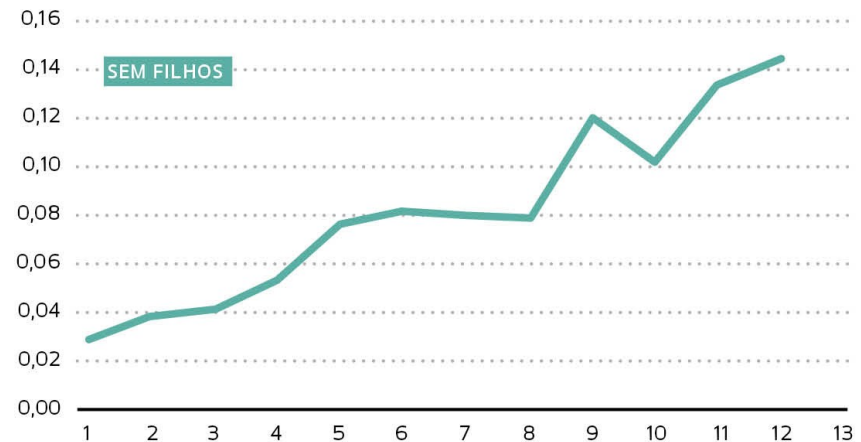
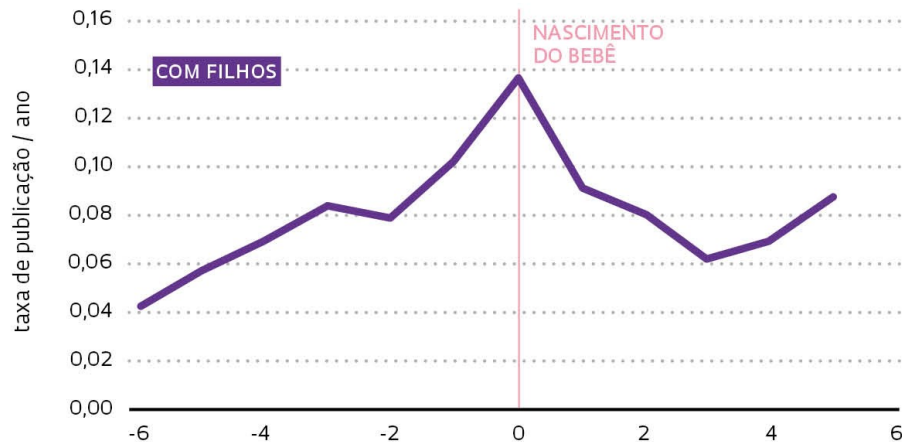
fernanda@fis.ufal.br



Por mais diversidade na ciência

Trabalho comprometido

Produtividade das pesquisadoras que se tornaram mães cai significativamente em relação às que não têm filhos



FONTE: PARENT IN SCIENCE

@neurocompbr.org
@soapboxsciencemaceio
@jedi_sbf

Neurociência
computacional:
neurocompbr.org

Por mais diversidade na ciência

1) Procurem/Sejam redes de apoio

2) Gastem um minuto a mais pensando “existe alguma mulher e/ou pessoa não-branca nessa área que eu poderia convidar?”

fernanda@fis.ufal.br

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@soapboxsciencemaceio

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